OPSC6312

POE Part 1- Research

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Introduction

In this project we are tasked to make an app of our own choosing. To decide what we want to make, we are required to research two apps that are similar to our general ideas. In this report I will be researching the Picture Insect and Picture Bird applications as they are similar to the type of app I was thinking about making. I will include screenshots of the apps and expand on my idea for an app.

Existing Application – Picture Insect

Overview

The app helps users identify insect species by analysing photos the user submits and matching it to similarly looking pictures in their database. They also appear to have an AI that helps identify insect species too. You upload a photo you've taken or take a new photo directly in the app. It tells you a lot of information about the bug in question which I generally find fun to know (Picture Insect - Bug Identifier, n.d.). It has had issues with a kind of "America-first bias" which extends to the greater northern hemisphere. What I mean by this is that it does not have as much skill identifying our local bugs and providing information pertaining to the southern African region as it does with the northern countries. This is to be expected in my opinion but does make one a little disappointed sometimes. I cannot seem to find any sources that explain the concept I speak of, if the affect is even a recognised event, but I am certain it is a thing (Oksana Skaldina and Blande, 2024, Queen Mary University of London, 2025, Wikipedia Contributors, 2019, Climate → Sustainability Directory, 2026).

Strengths

- Easy to understand and use interface
- Gives a good amount of information about insect
- Generally reliable
- Can be used for free without the everyman feeling like they are missing out
- Allows for knowledgeable users to correct misidentifications

Weaknesses

- Has a tendency to suggest northern hemisphere bugs, could use a geographical something to let users filter suggestions by geographical region
- Asks me to by premium every time I open the app, little annoying

Implementation

I did attempt to contact the Picture Insect team to ask about the development directly, but they understandably were not able to share any of the details about the development process and peripheral activities. If I had to guess, I would say the app was made separate from the "AI" as it stands today, as I am nearly sure it did exist prior to our current definition of AI. From the details they *did* share in the email, they have a team of designers as well as the development team, which would indicate to me that there was a lengthy design process. They did mention that their internal tools were proprietary, which could indicate that it was not created solely with Android studio, if it was at all. It was most likely developed in an agile method, building one feature at a time then testing that feature as well as checking that the existing features haven't broken. It uses AI as mentioned before, so perhaps an API for their AI? There also seems to be a team of experts employed, so there was probably rounds of testing where the AI would guess the bug and then the actual expert would verify it.

Screenshots



This is the first screen you are greeted with when opening the app. It advertises the premium service, but after using it a while the casual user will realize they may not need premium.

Figure 1 Picture Insect first screen

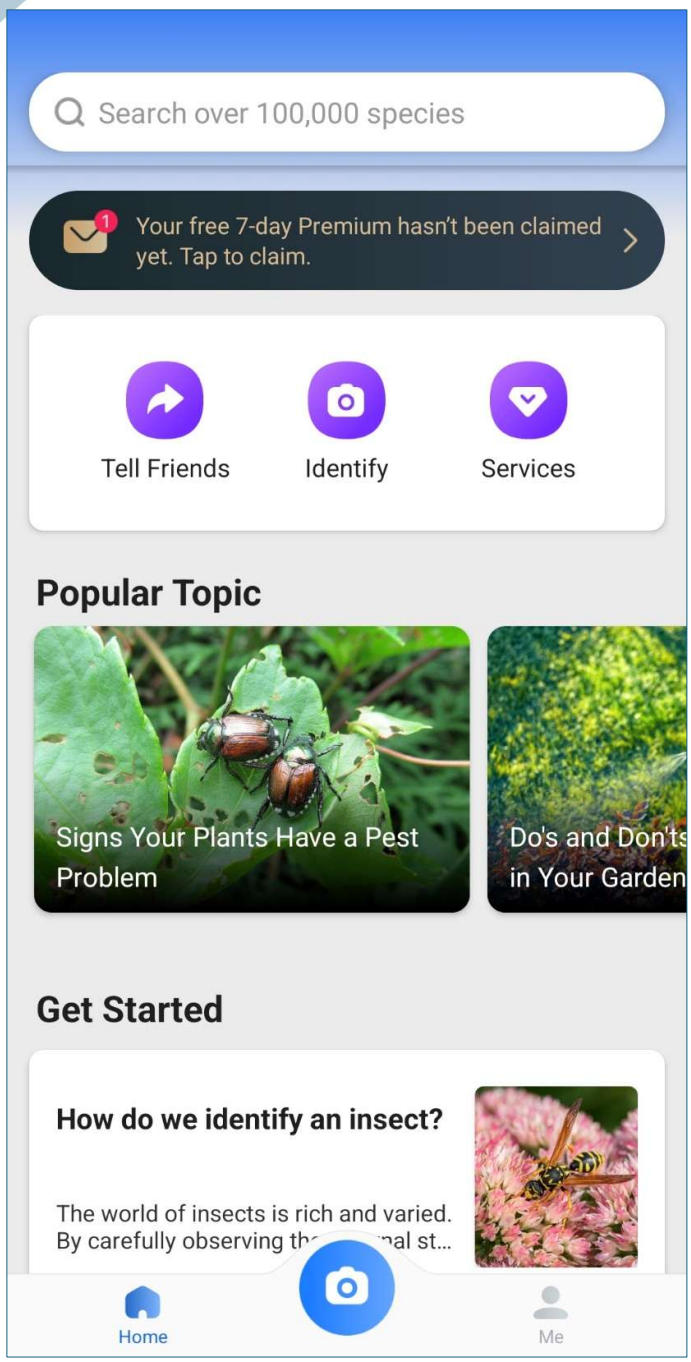


Figure 2 Picture Insect Home screen

This is the "Home screen" of the app, it has several functions. It lets you search for a bug, open the camera to use the image identification feature and offers you articles to read about various bug related topics.

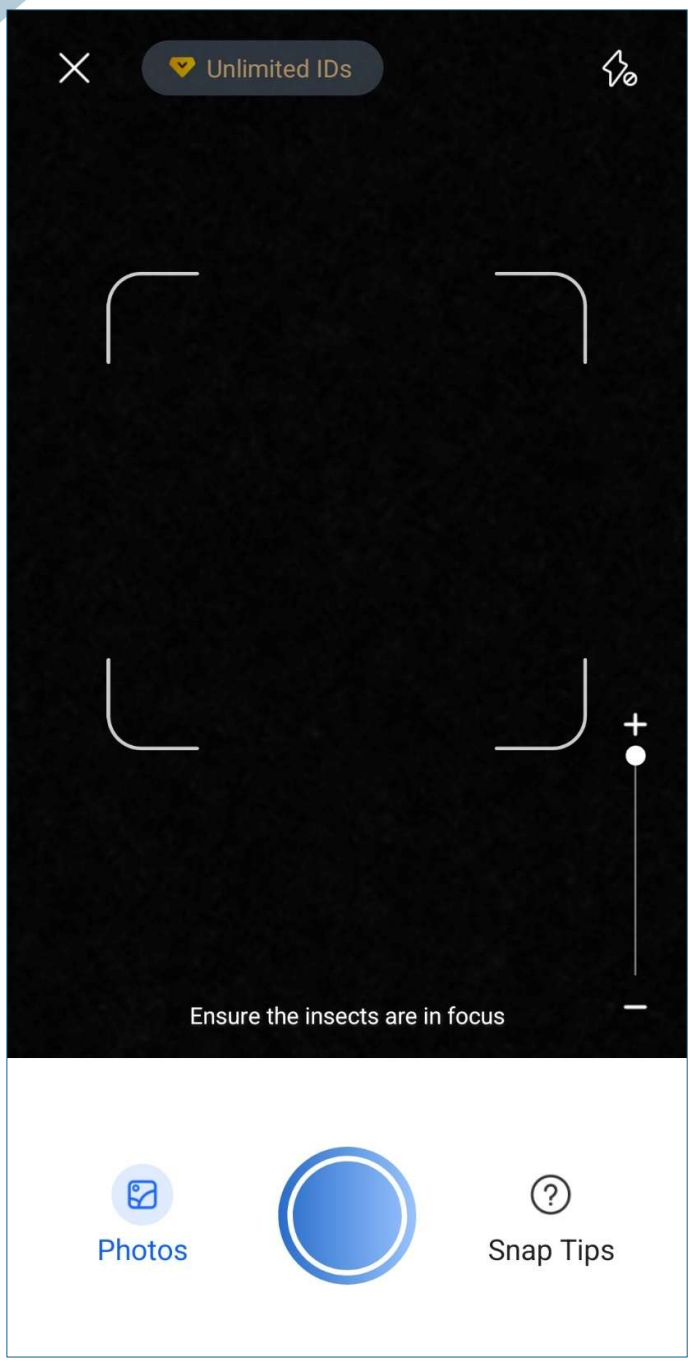


Figure 3 Picture Insect Camera UI

This is the camera UI, where you can take pictures of the bug you've seen. You can also select photos from your phone that you've taken previously. It includes a zoom feature and a flash control, as well as a box to show you where to position the bug.



Figure 4 Picture Insect Image adjustment screen

This is the UI once you've taken/selected the desired photo, it lets you reposition the bug and zoom in for a better identification chance. The text and UI elements can sometimes be lost on some image colours (unsure if it is dynamically colour changing).

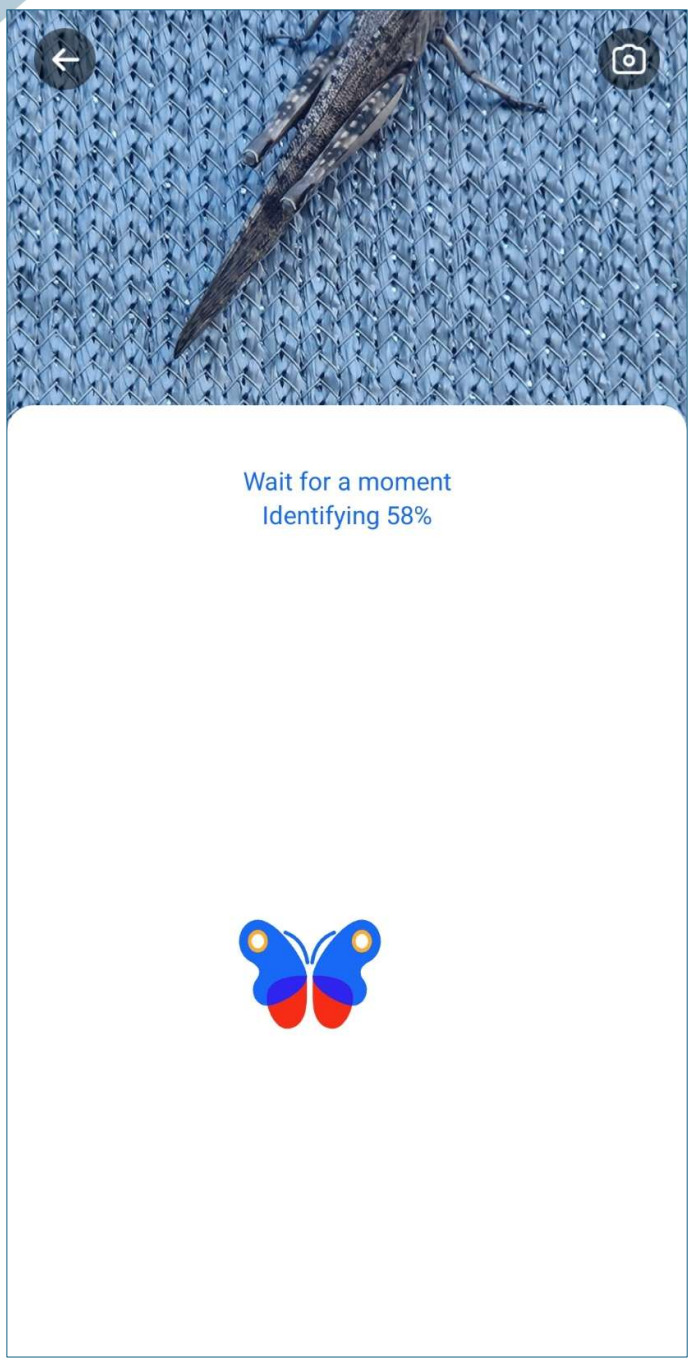


Figure 5 Picture Insect Analyzing screen

This is the in-between screen for the app to identify possible candidates once you've submitted your photo. It visually shows the user that it is doing something and pretty decently shows that that something is it scanning the image to identify the bug.

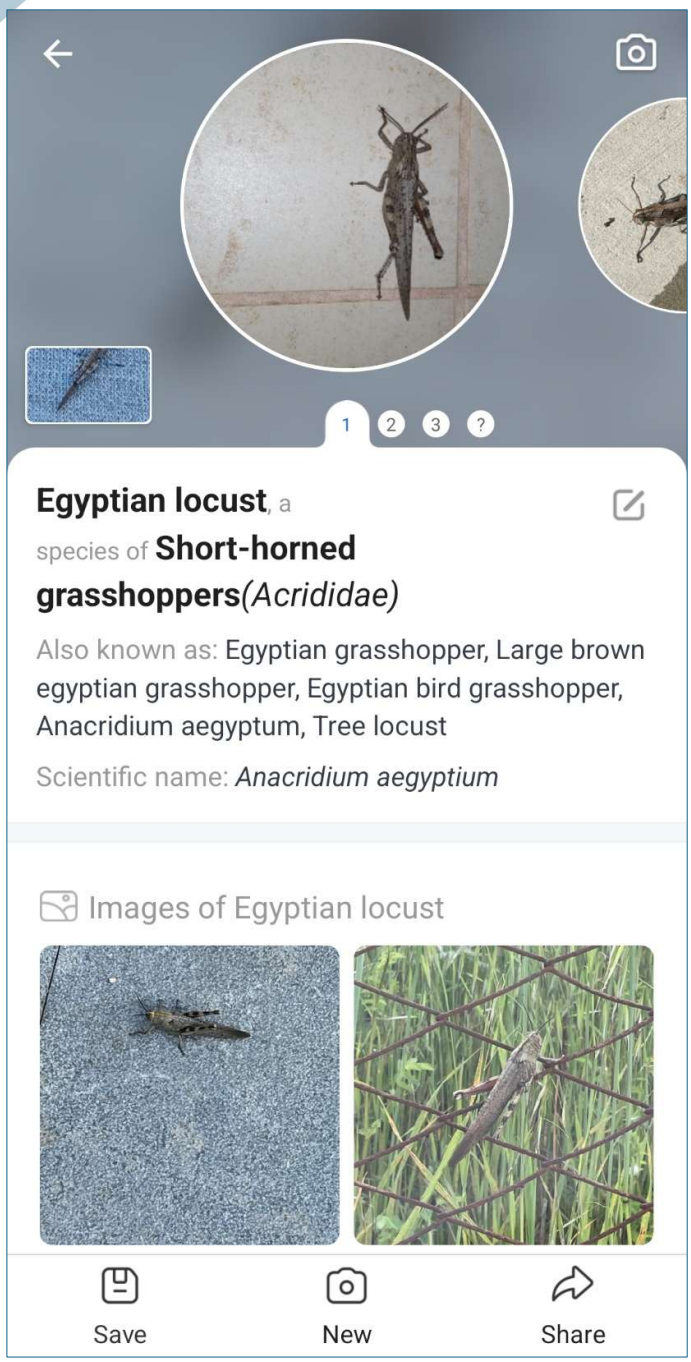


Figure 6 Picture Insect Possible identity screen

This is the UI for when it serves the user the options that it thinks are closest to the bug in the photo. It has 3 guesses to offer, with 1 being the best guess it has. It shows you different names and the scientific name as well. It contains pictures of the bug for you to compare yours against.

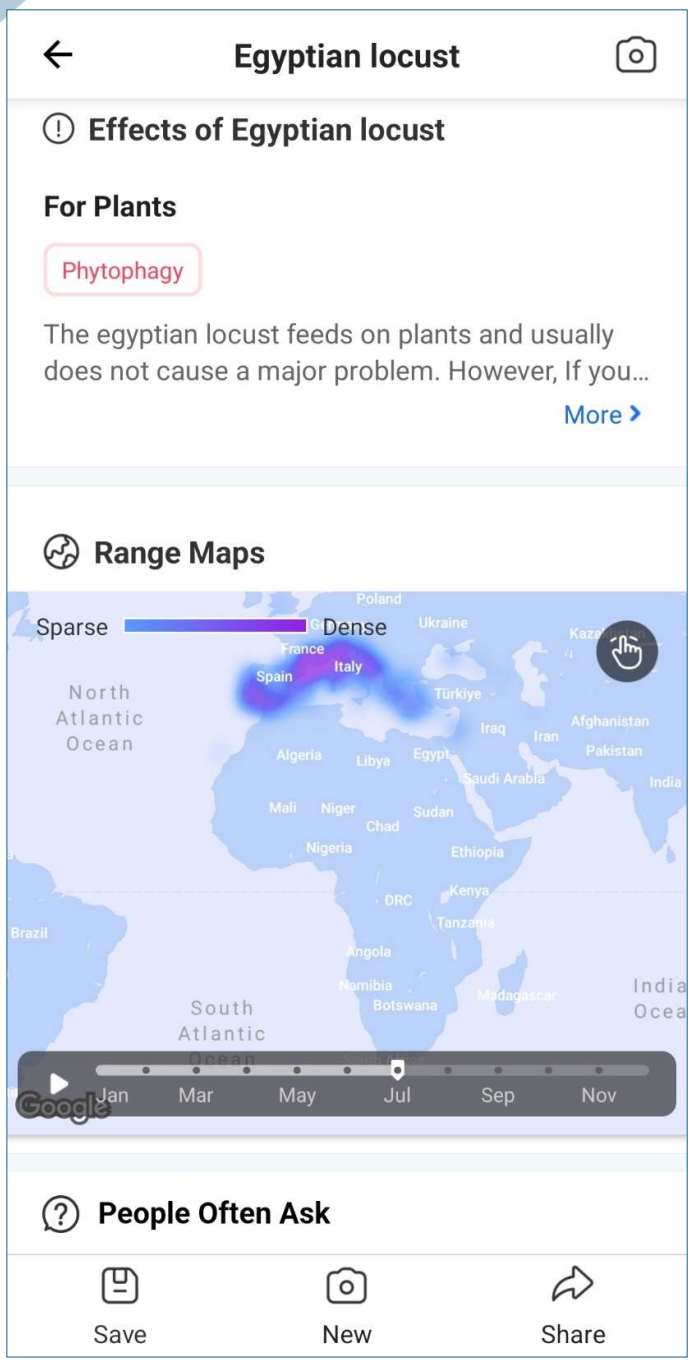


Figure 7 Picture Insect Identity continued

If you scroll down you will see more information about the range and problems of the bug. As you can see, it has little to no data on the African continent despite me seeing this particular bug right here in Cape Town.

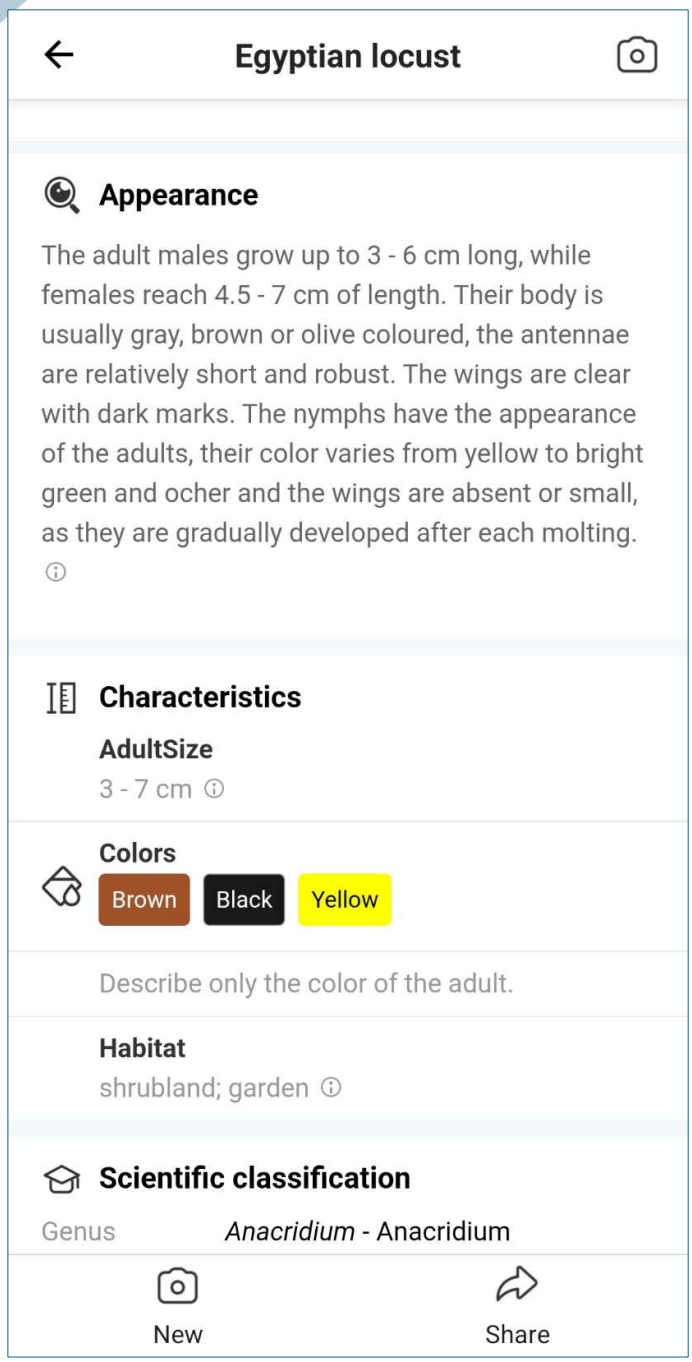


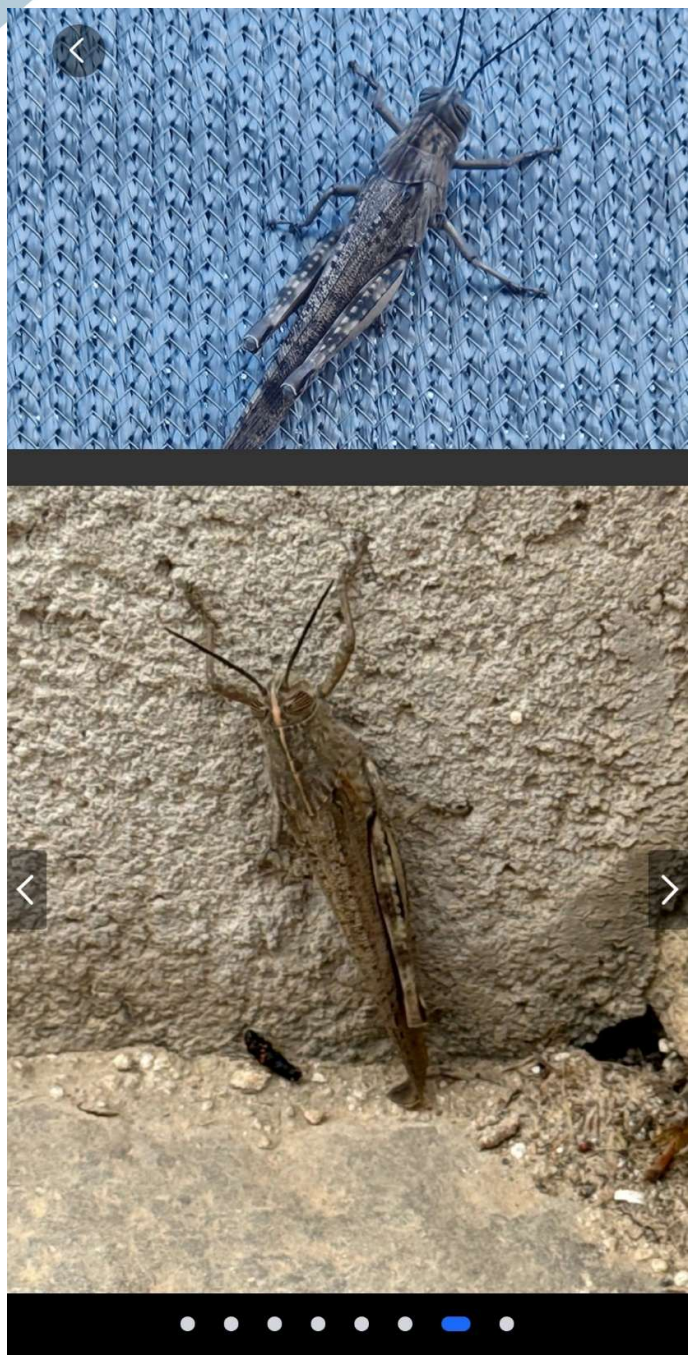
Figure 8 Picture Insect Appearance and characteristics information

Scrolling further, you are shown a description of the bug as well as basic characteristics. This is very useful to me as size can often be a good thing to use to verify the species. I have had times where the insect looked similar but the size was far from correct.



Figure 9 Picture Insect Wikipedia page display

If you were to tap the round icon with a ? in the centre that is seen in the above image at the end of the description as well as appearing in the characteristics, it will take you to the Wikipedia page for said bug, which is a very cool feature.



It has a feature that lets you directly compare your image with the stored images of the likely candidate so you can judge for yourself if it is the correct bug. I find this feature really nice.

Figure 10 Picture Insect Image comparison screen

Existing Application – Picture Bird

Overview

This application is similar to the aforementioned application, as it provides many of the same services. The bird app includes some extra features however, including the ability to use audio recordings of bird song to identify bird species. It still includes the bias discussed above, but to a lesser extent with photo identification; the audio identification does suffer from the bias quite a bit however (Oksana Skaldina and Blande, 2024, Queen Mary University of London, 2025, Wikipedia Contributors, 2019, Climate → Sustainability Directory, 2026). It does contain a map feature that is similar to the application that I want to create, except mine will be for insects instead of birds. It lets you see likely bird species near you and will show how many reports there have been recently. It also seems like it contains a function to guide you to the location too, which is an interesting idea. I must admit when I came up with the idea for my app, I had not known that it had essentially already been half done. While mine is a lot less sophisticated, I think it is still a valid idea (Picture Bird - Online bird encyclopedia and bird identifier, n.d.).

Strengths

- Easy to understand and use interface
- Gives a good amount of information about the bird
- Generally reliable
- Can be used for free without the everyman feeling like they are missing out
- Allows for knowledgeable users to correct misidentifications
- Let's users use audio to identify birds they cannot see but can clearly hear

Weaknesses

- Has a tendency to suggest northern hemisphere birds (more with the audio identification feature), could use a geographical something to let users filter suggestions by geographical region
- Asks me to by premium every time I open the app, little annoying

Implementation

I did attempt to contact the Picture Bird team to ask about the development directly, but they understandably were not able to share any of the details about the development process and peripheral activities. If I had to guess, I would say the app was made separate from the "AI" as it stands today, as I am nearly sure it did exist prior to our current definition of AI. From the details they *did* share in the email, they have a team of designers as well as the development team, which would indicate to me that there was a lengthy design process. They did mention that their internal tools were proprietary, which could indicate that it was not created solely with Android studio, if it was at all. It was most likely developed in an agile method, building one feature at a time then testing that feature as well as checking that the existing features haven't broken. It uses AI as mentioned before, so perhaps an API for their AI?

The map functionality probably uses a map API, something I will need to also use.

Screenshots

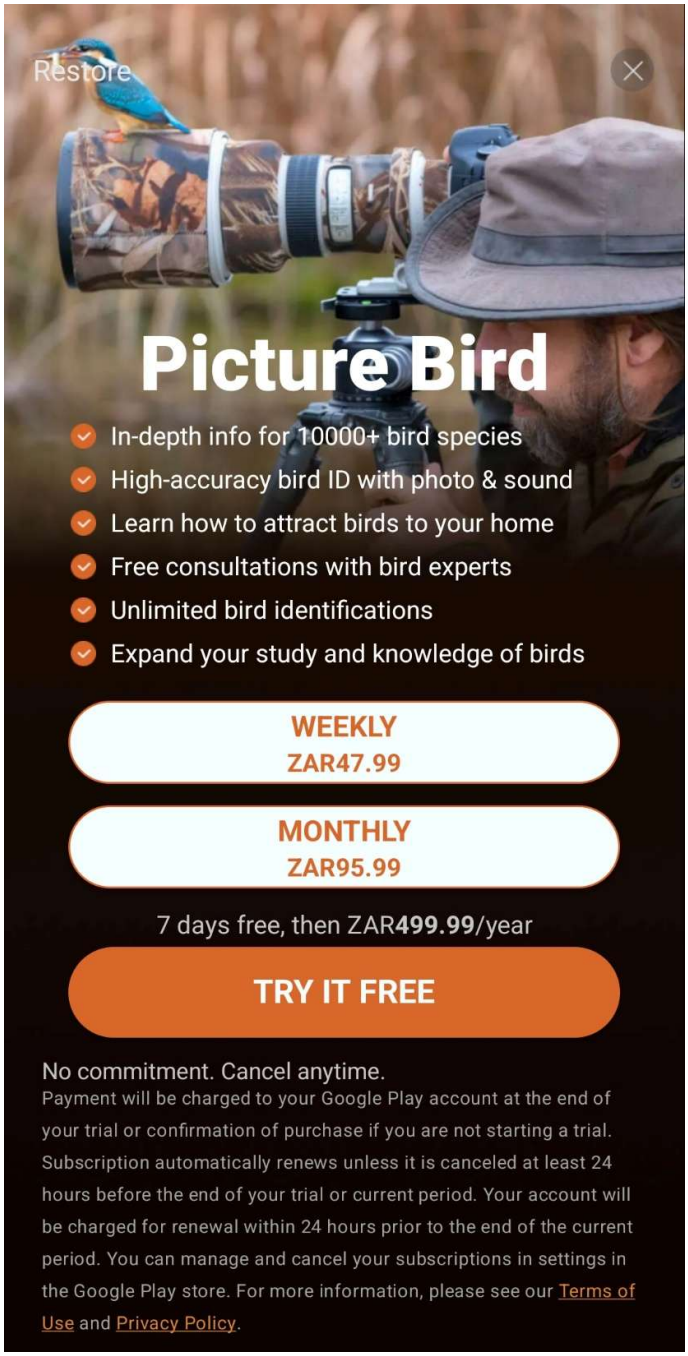


Figure 11 Picture Bird First screen

This is the first screen you are greeted with when opening the app. It advertises the premium service, but after using it a while the casual user will realize they may not need premium.

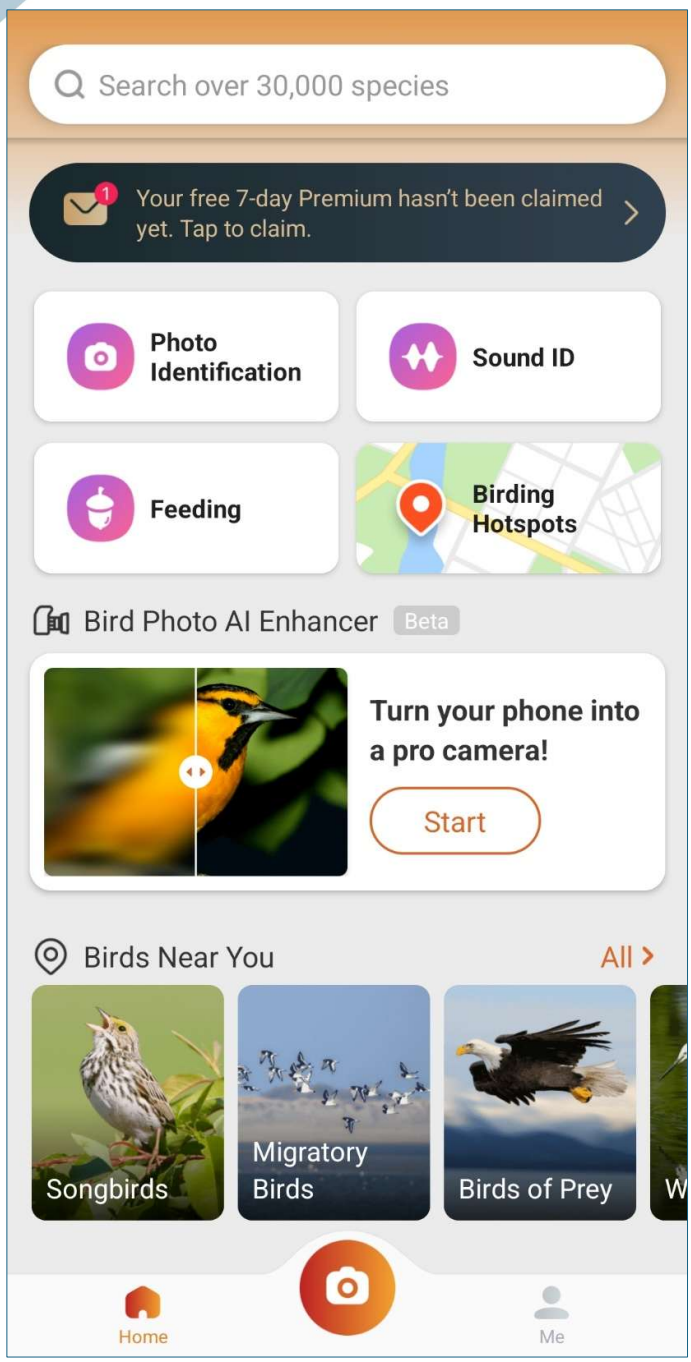


Figure 12 Picture Bird Home Screen

This is the "Home screen" of the app, it has several functions. It lets you search for a bird, open the camera to use the image identification feature, open an audio capturing functionality to identify birds by their song, a page for tips on feeding birds, birding hotspots and other features.

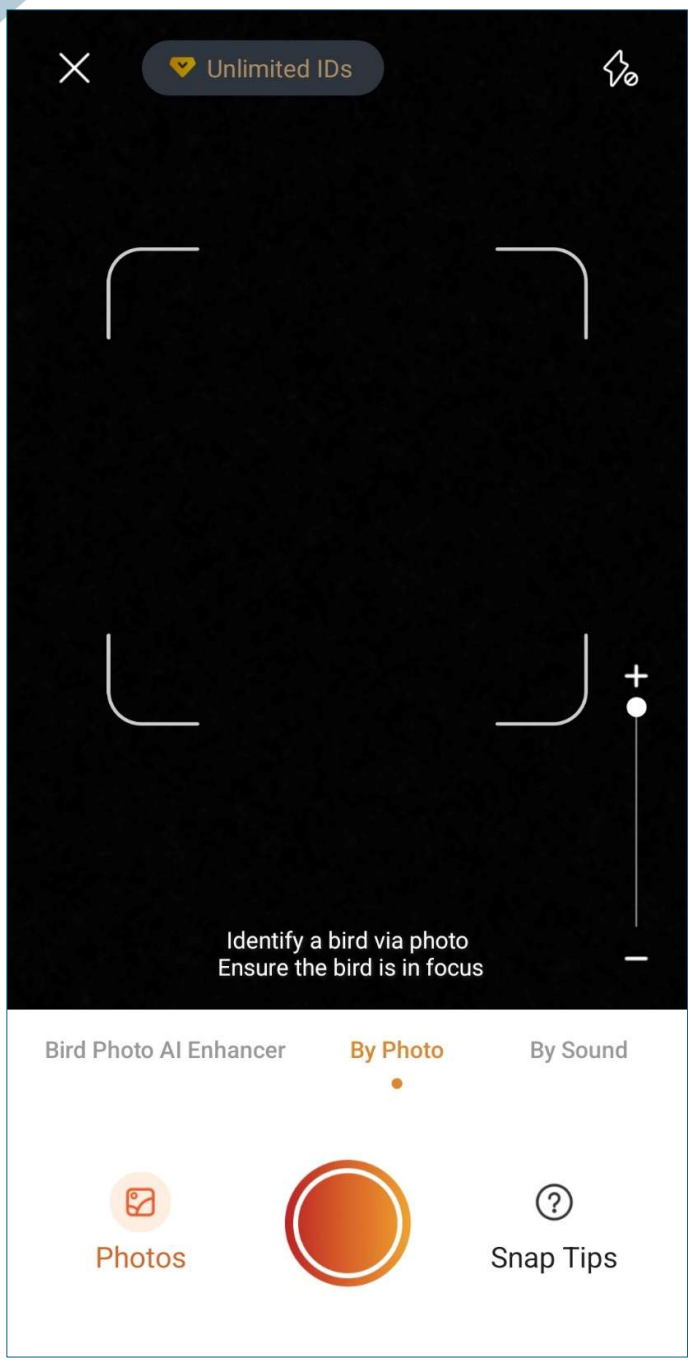
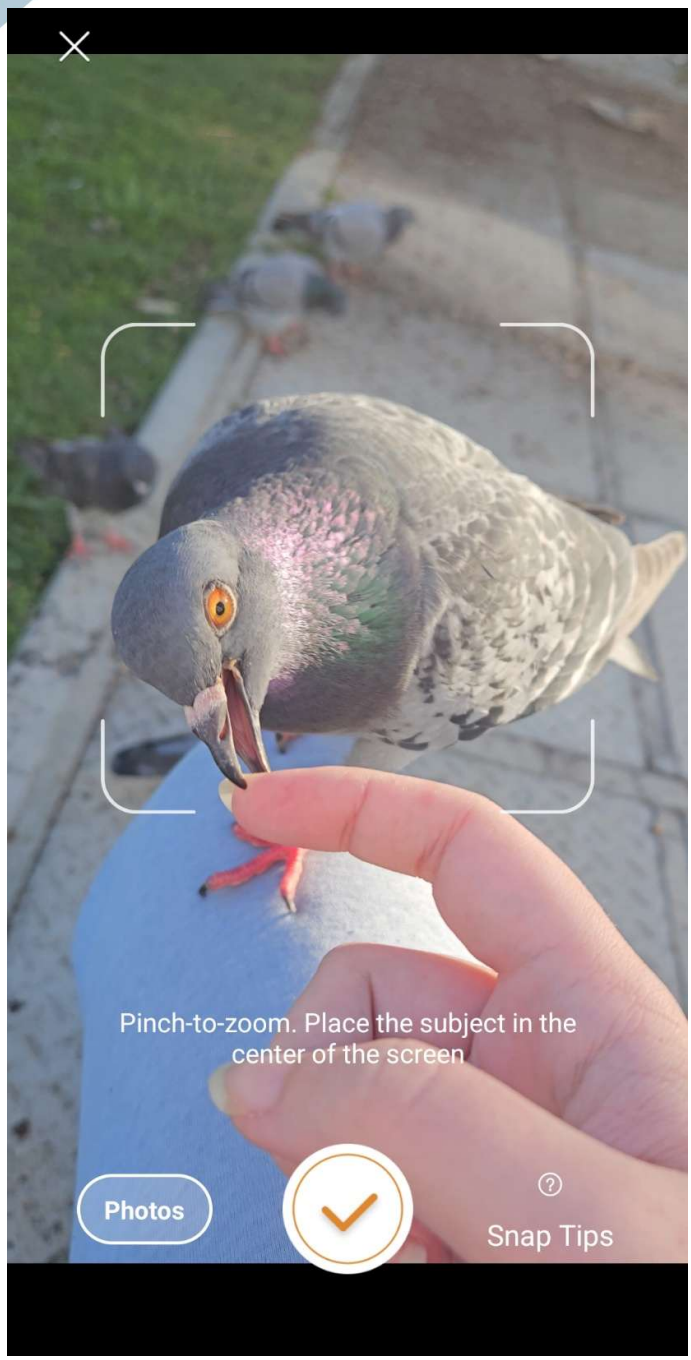


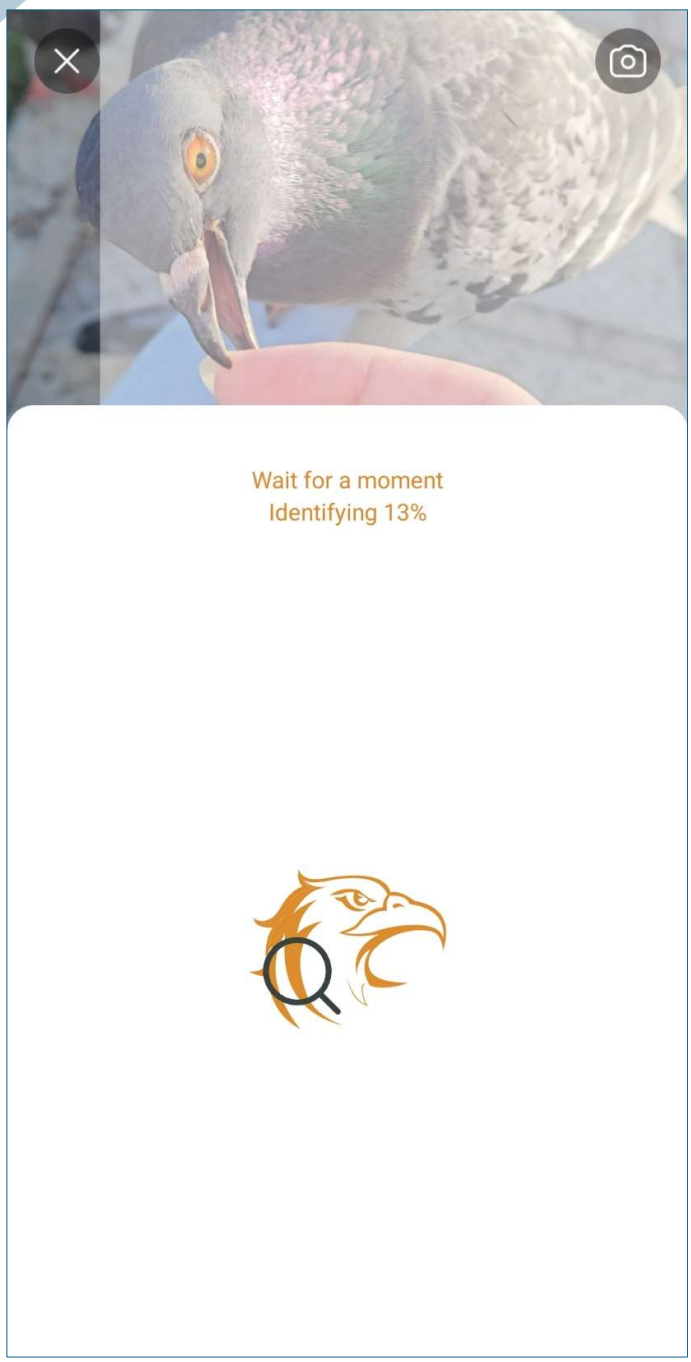
Figure 13 Picture Bird Camera interface

This is the camera UI that lets you take pictures or upload your own images for analysis. It contains the same UI elements as the Picture Insect UI, but with the extra options to hotswap to the AI image enhancer or the sound ID function.



The UI for framing the image correctly for optimal identification chances.

Figure 14 Picture Bird Photo adjusting screen



The in-between screen where the app does its thing and analyses the image and attempts to find a match.

Figure 15 Picture Bird Analysing screen

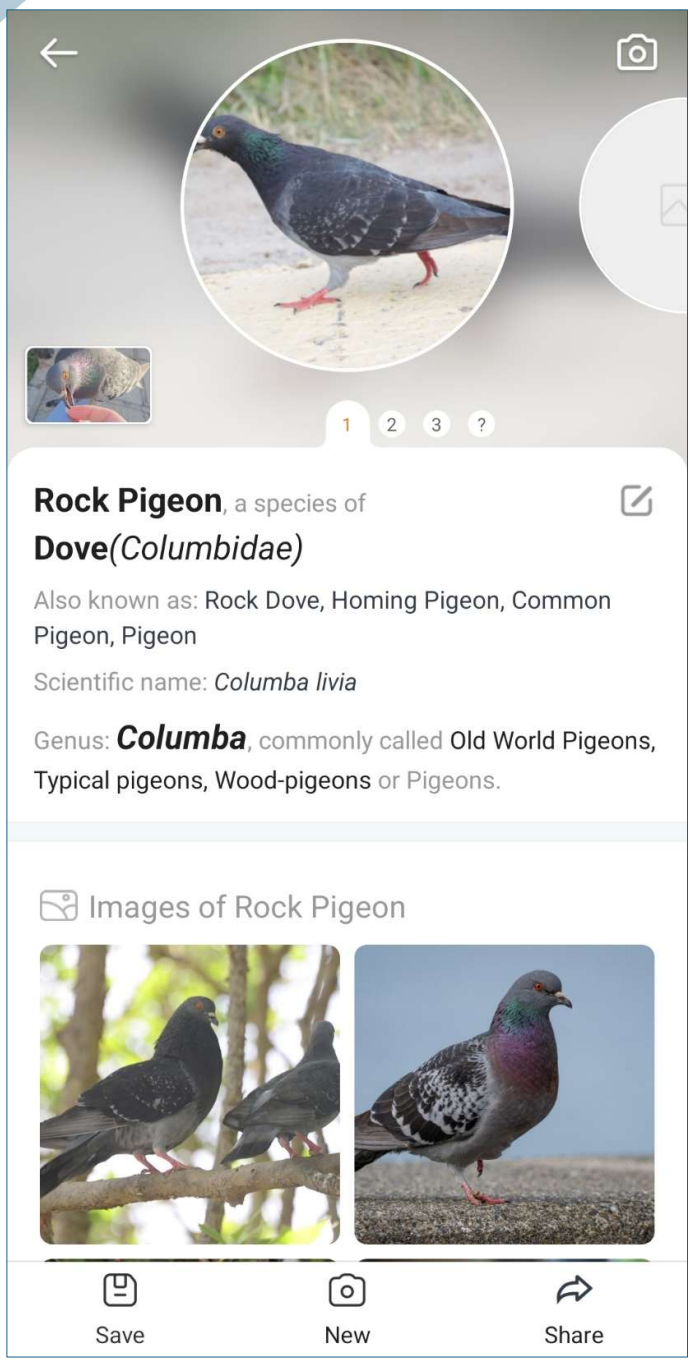
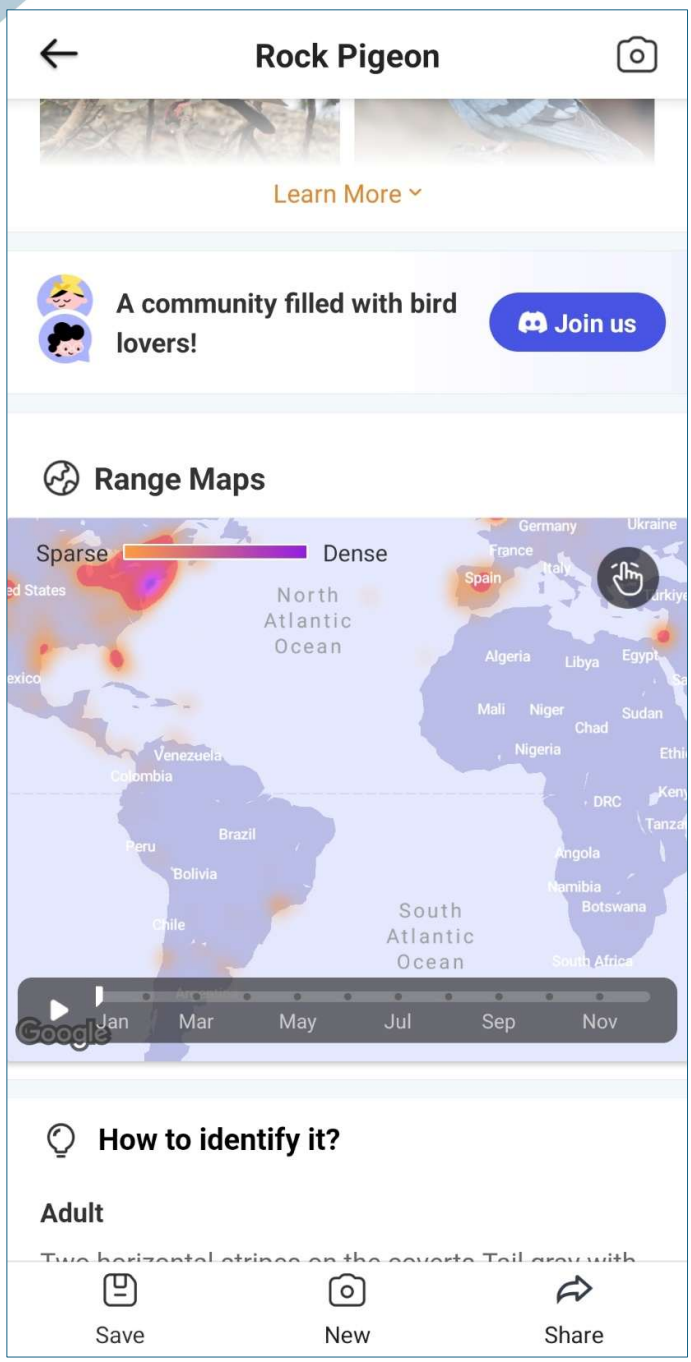


Figure 16 Picture Bird Possible identity screen

This is the UI for when it serves the user the options that it thinks are closest to the bird in the photo. It has 3 guesses to offer, with 1 being the best guess it has. It shows you different names and the scientific name as well. It contains pictures of the bird for you to compare yours against.



Here we see the range map with a species that is very widespread.

Figure 17 Picture Bird Identity screen continued

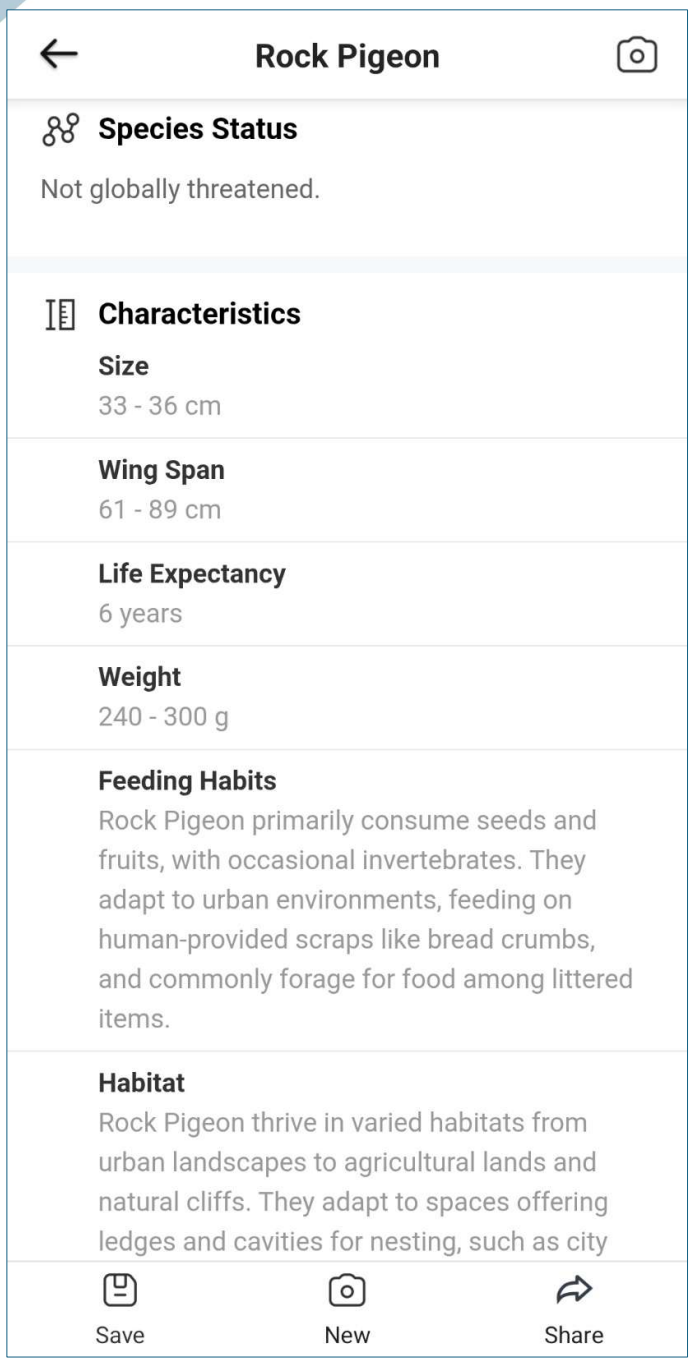


Figure 18 Picture Bird Characteristics continued

This app also shows general characteristics of the identified subject.

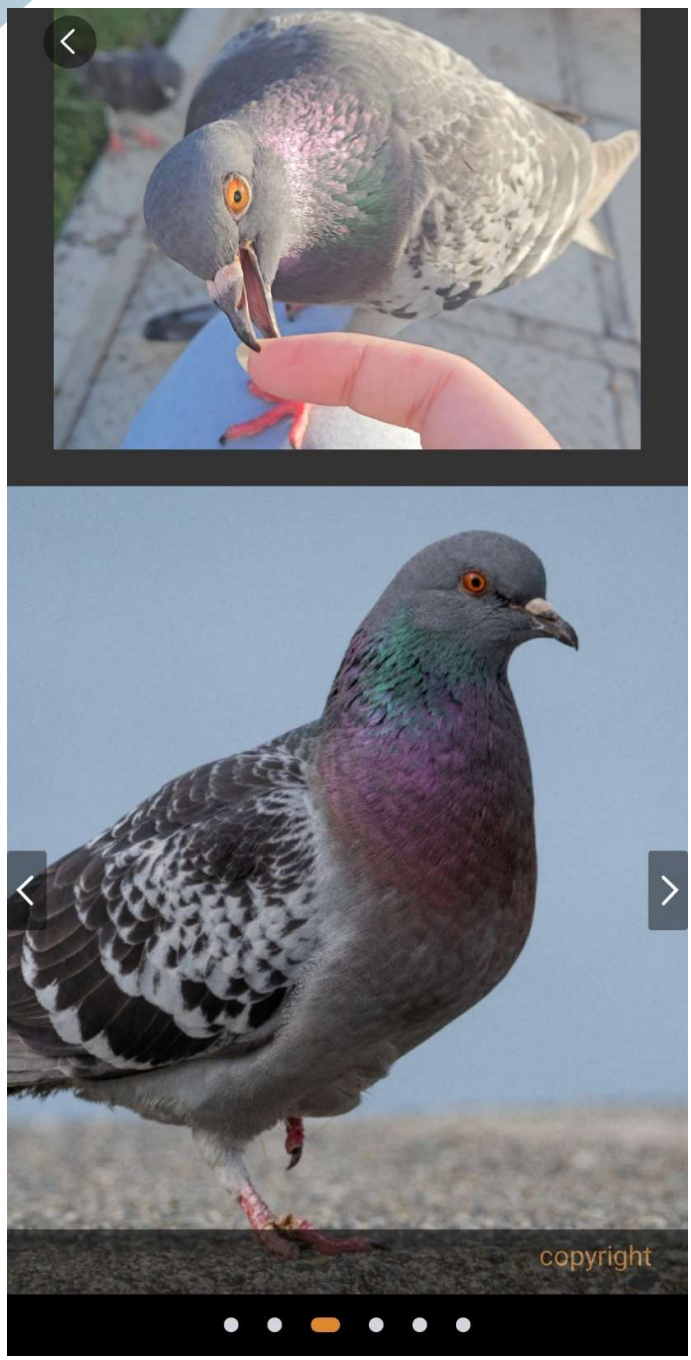


Figure 19 Picture Bird Image comparison

The image comparison may look a little silly with the example image used, but for far away birds or LBJs it comes in handy. Especially if you have a shot of the underwings only and need to compare them.

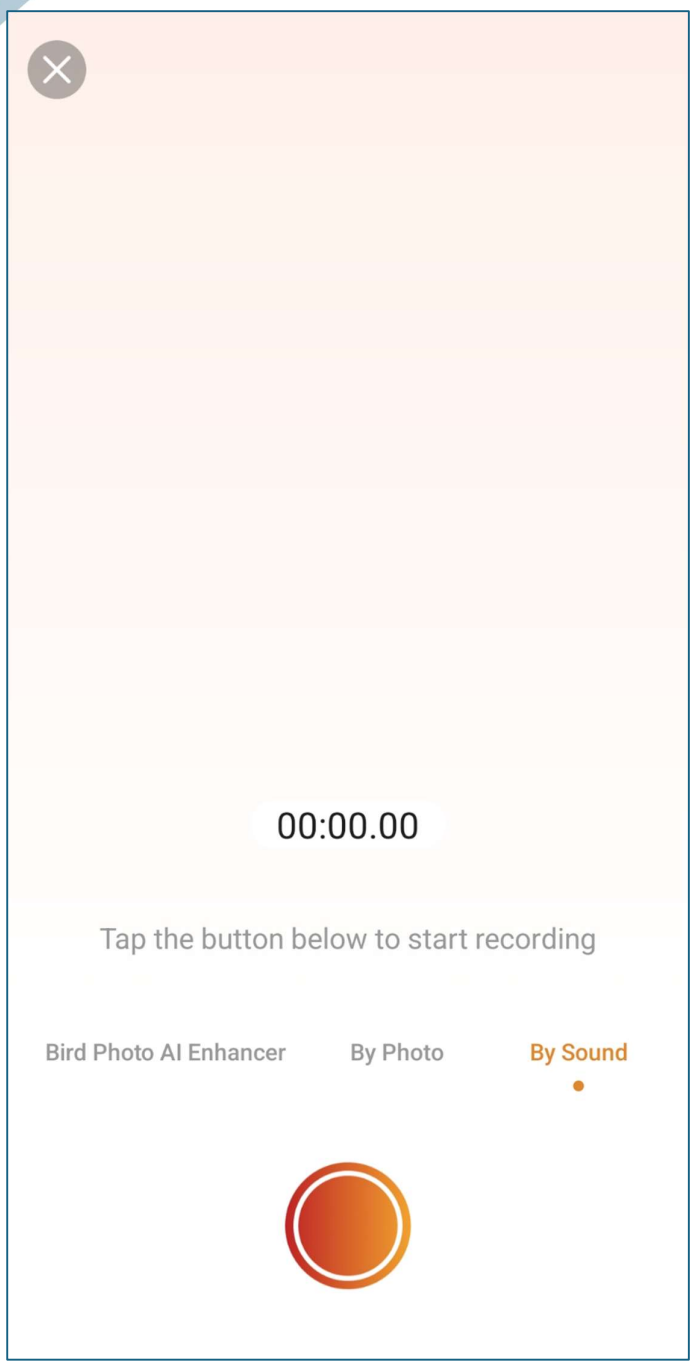


Figure 20 Picture Bird Audio UI

This is the interface you see when you select the audio option. It is simple and I think it benefits from that simplicity. It does not let you upload your own audio from a file, you must record it live through the app.

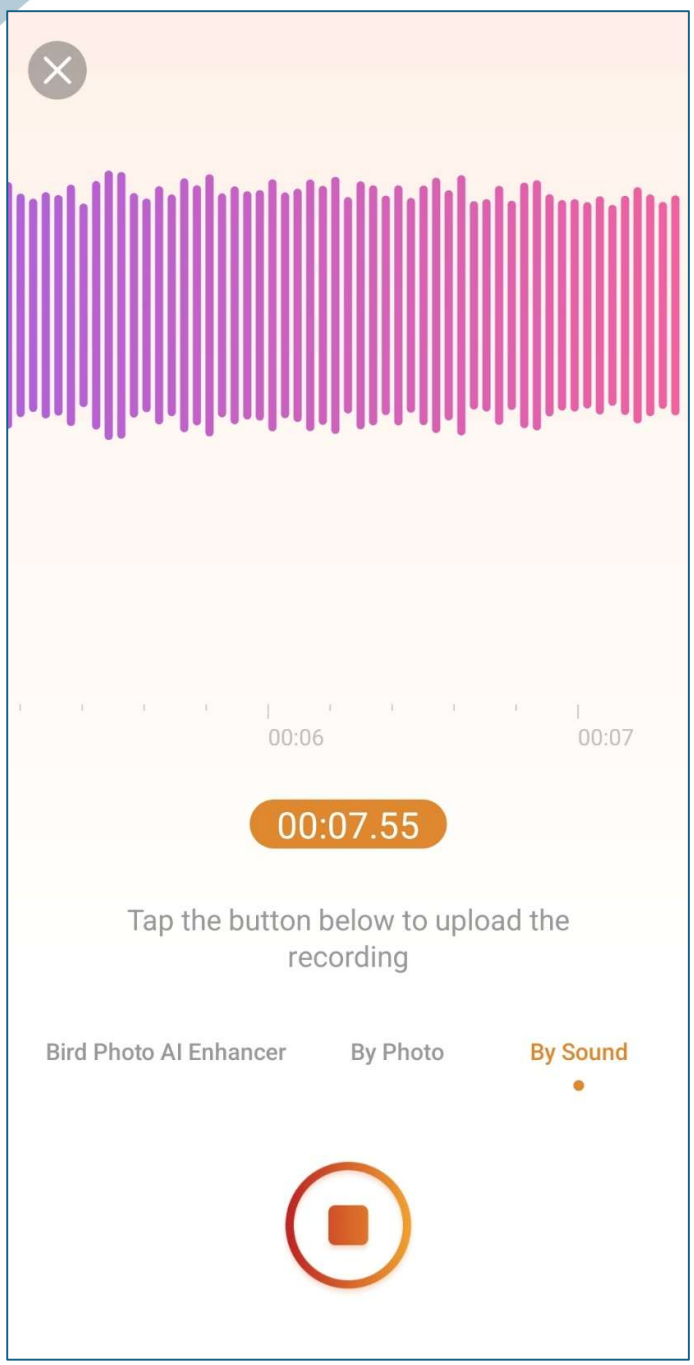


Figure 21 Picture Bird Recording audio interface

This is the UI you see when you are recording the sound you're hearing. It does tell you to get at least 5 seconds of sound to help them identify the call. Again the interface is simple and I think it benefits from it.

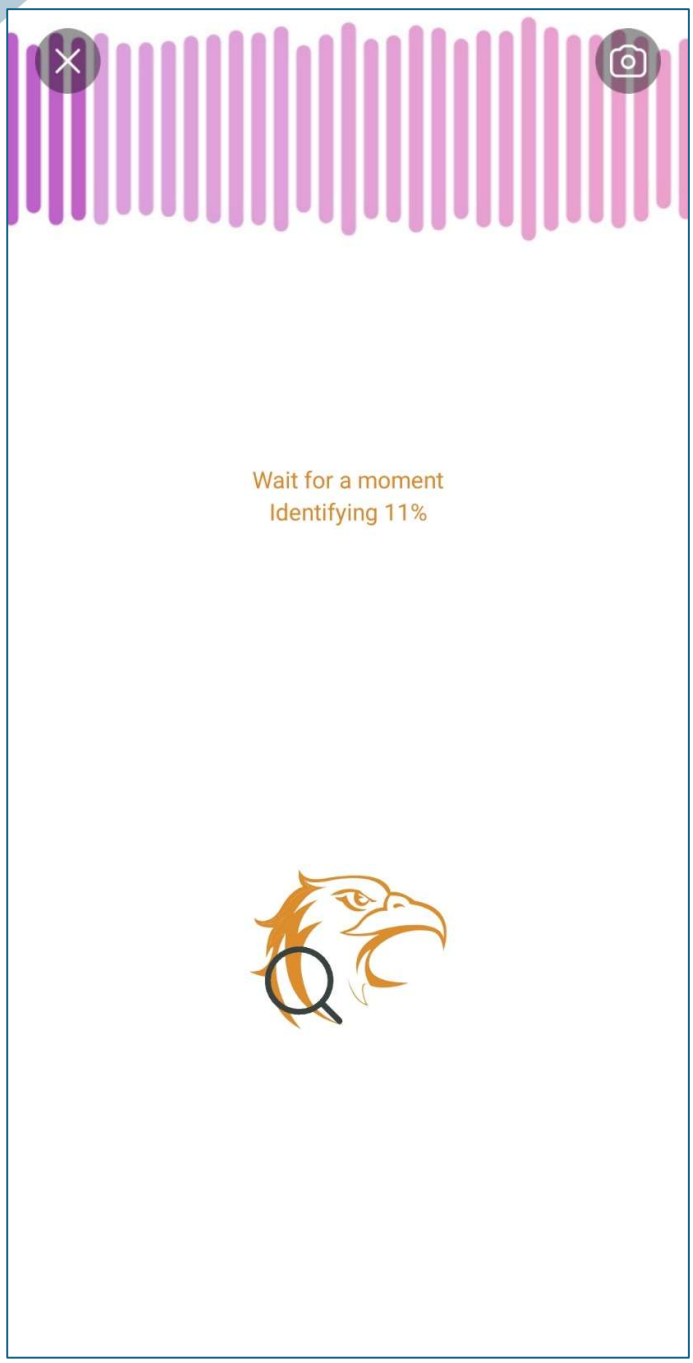


Figure 22 Picture Bird Result loading

The identifying loading screen is similar to the image one. It works to make the user feel that the program is busy working and so you don't really mind waiting.

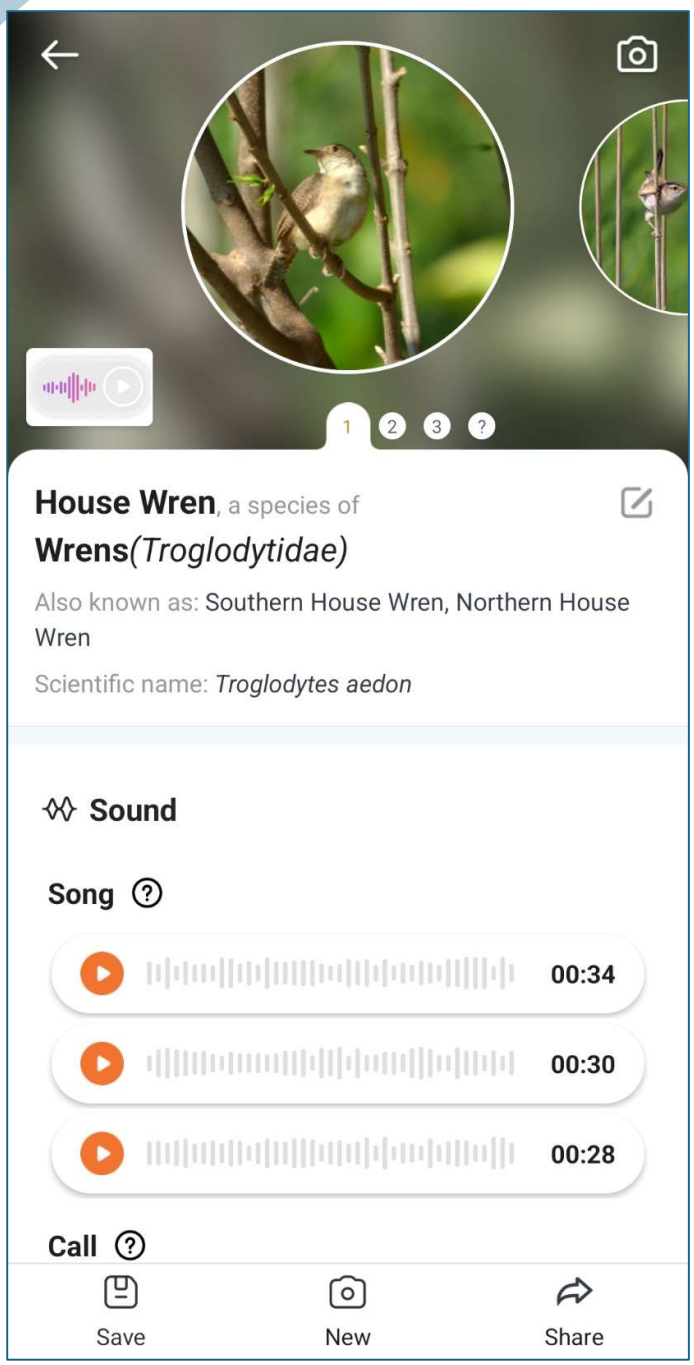


Figure 23 Picture Bird Audio results screen

The results page contains most of the same stuff as the image results does, though the audio one has the birds calls for you to play and compare to yours. Unfortunately the result is incorrect, but it gives me a good opportunity to show what the procedure is to correct an identification.

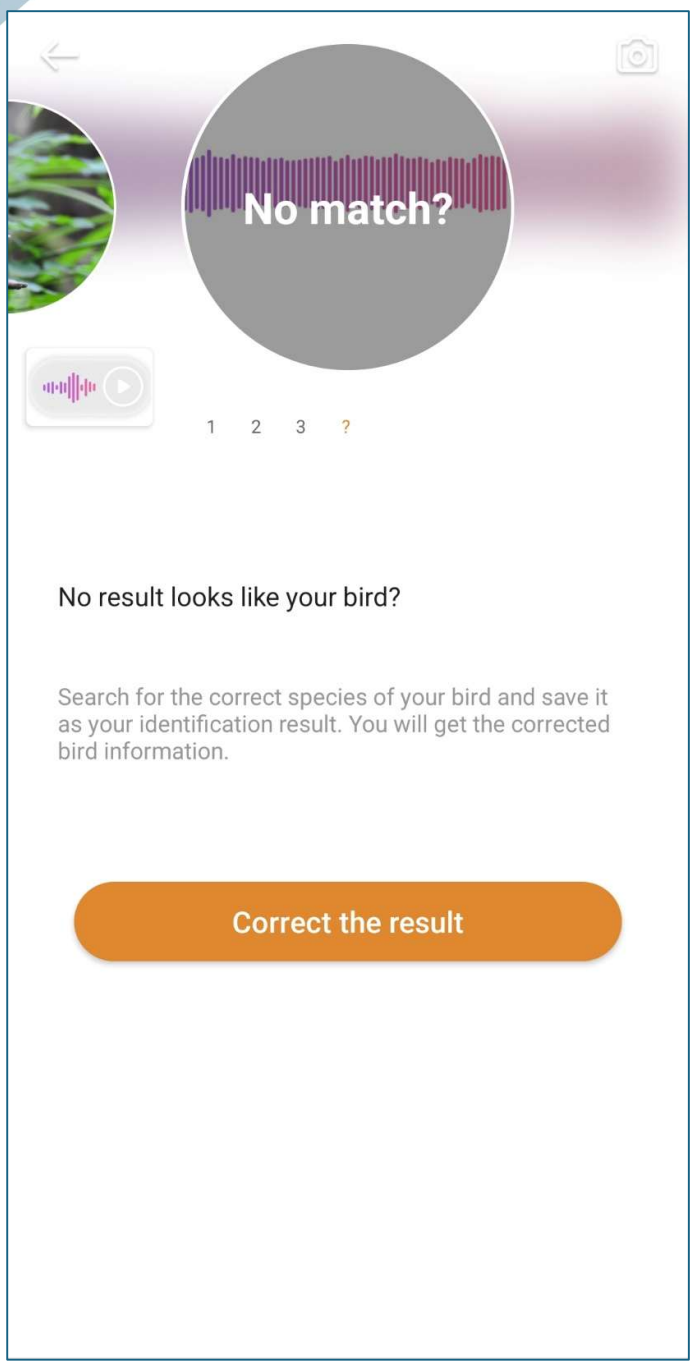


Figure 24 Picture Bird No match screen

This is the screen if you swipe through the results and decide that their guesses are incorrect, and that you are pretty sure what it actually is. You can correct the result yourself which is a useful feature for both the single user and the wider user base, as it will improve the identification AI(?) by correcting it.

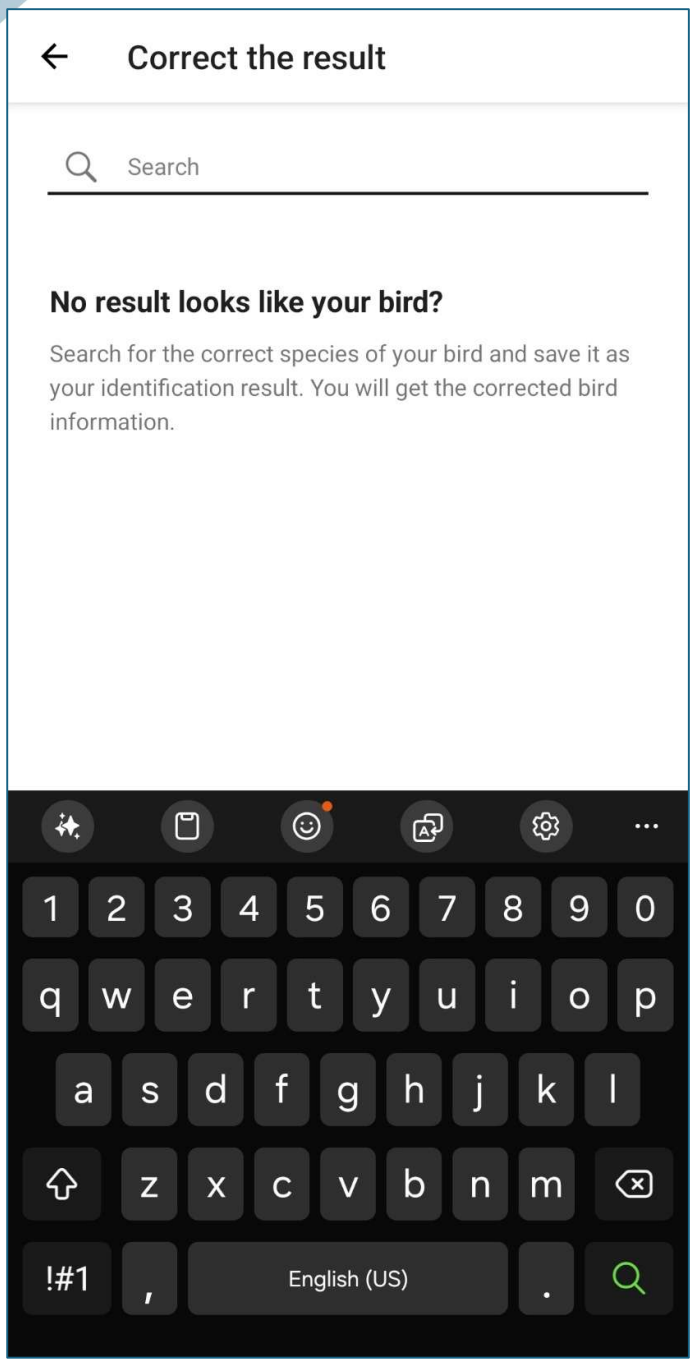


Figure 25 Picture Bird Correct the result interface

When you correct the result, you are asked to type the name of the bird so that it can search their database of birds to add this information to that birds file.

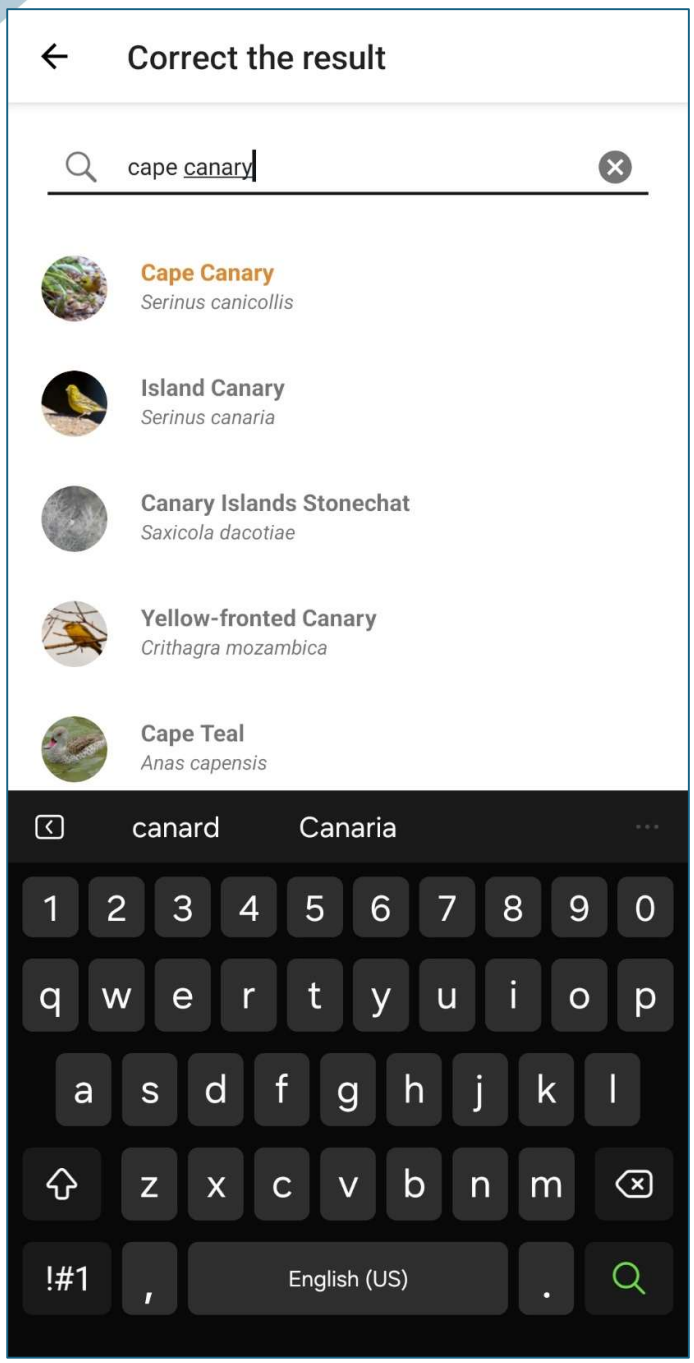


Figure 26 Picture Bird Suggested birds based on typing

When typing the name of the bird, the results will auto populate as you type so make your matching faster.

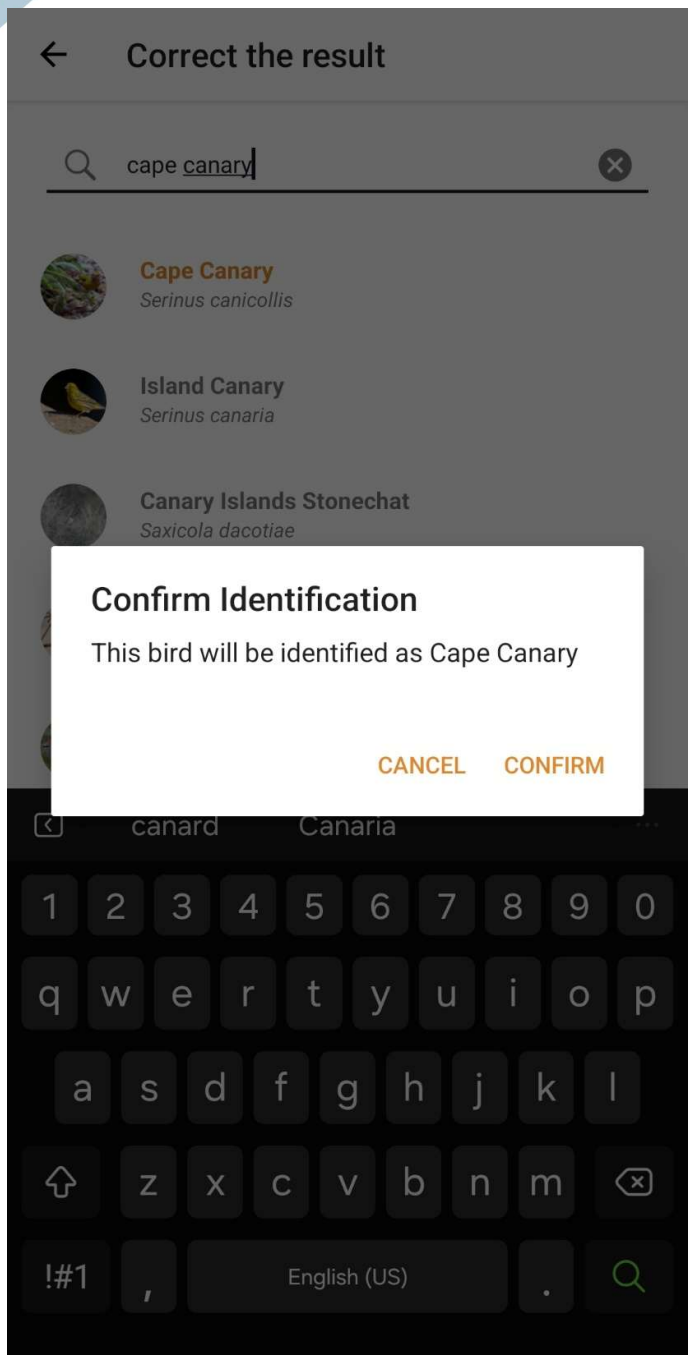


Figure 27 Picture Bird Confirm result pop up

It will ask you to confirm that that is the species you want to identify your image/audio file as before proceeding. It is a nice way to avoid accidental picks and to make sure the user chose the correct one.

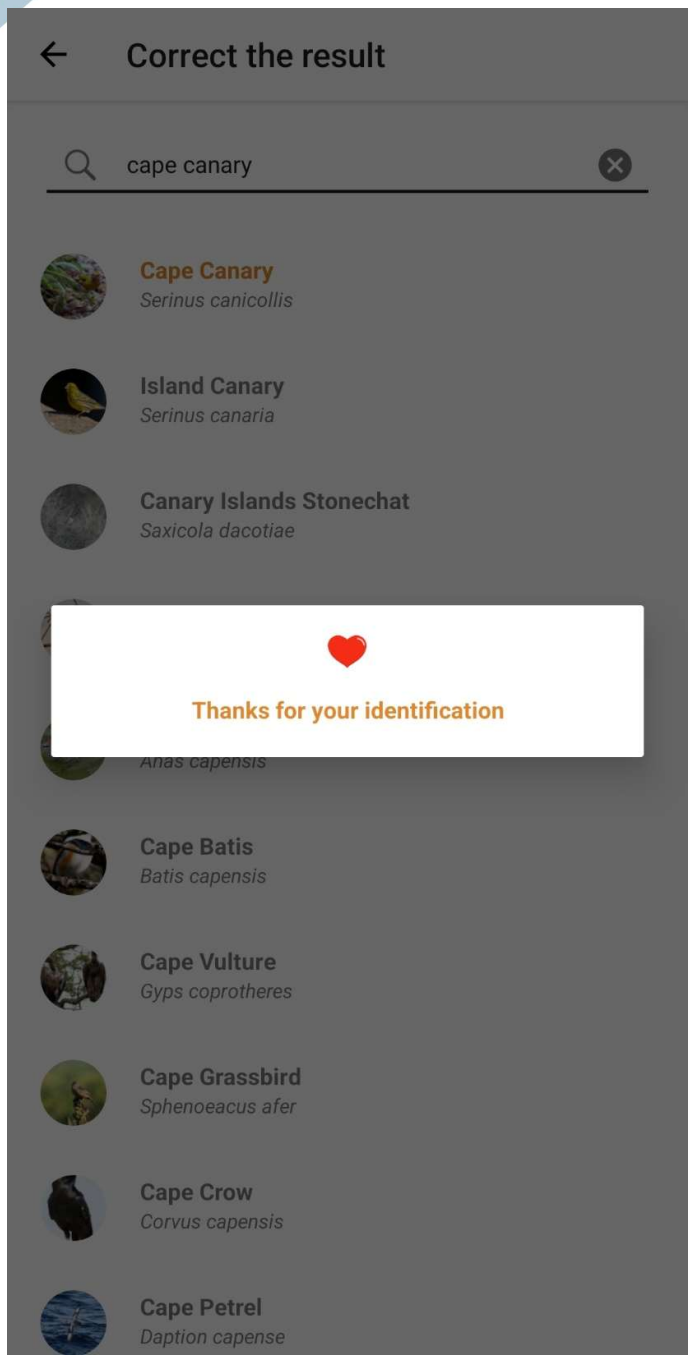


Figure 28 Picture Bird Thanks for ID pop up

When you confirm an ID, it will thank you for your identification, which I think is a nice feature.

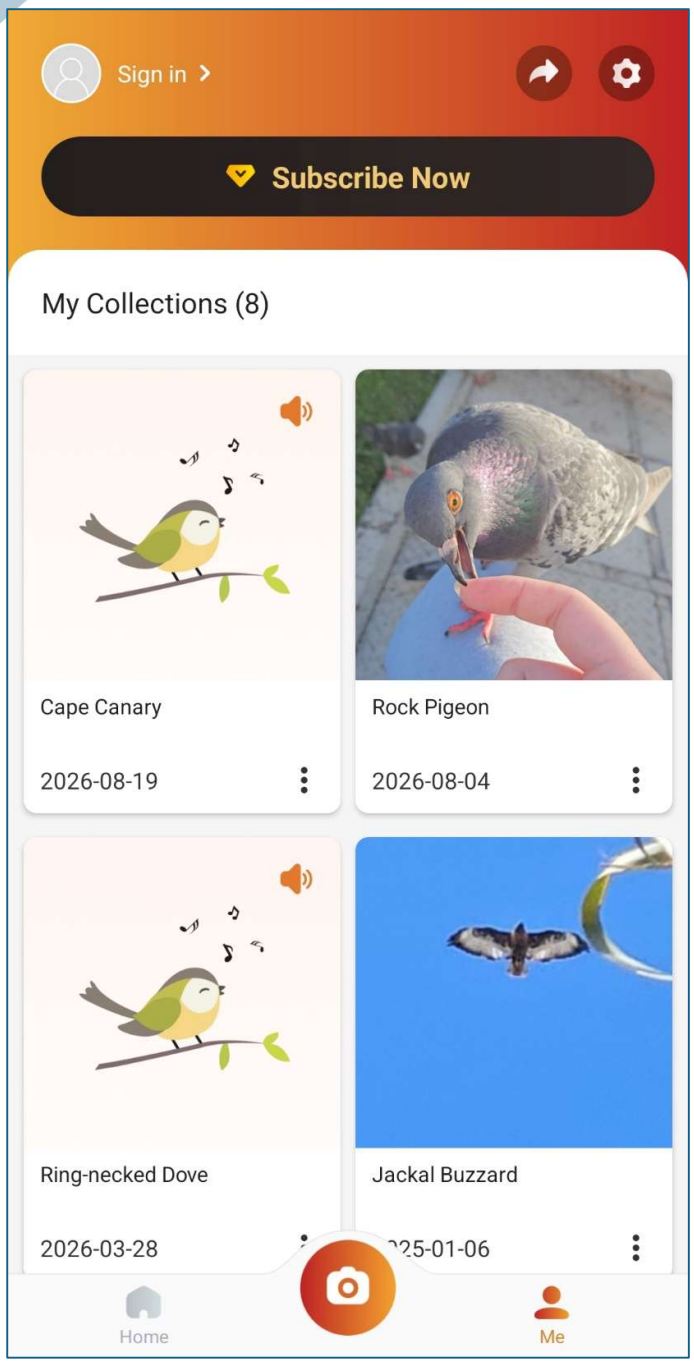


Figure 29 Picture Bird Previous ID page

This is the page that shows all your previous identifications. It is useful to look back on your birding history.

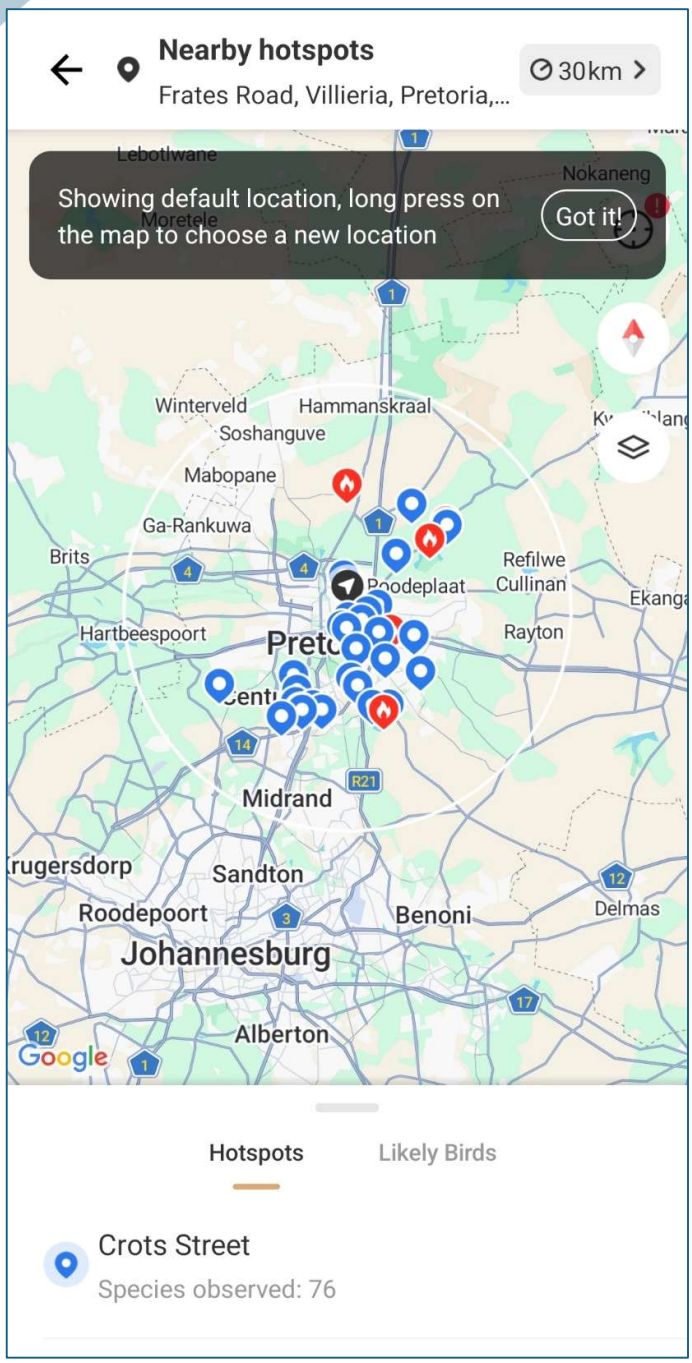


Figure 30 Picture Bird Map UI

This is the Map UI on opening. The default location is apparently in Pretoria, and since I haven't given my location to the app (I am very selective with that information) it won't auto suggest based on my actual location. There is a lot to look at here, and we will go over them on further screenshots.

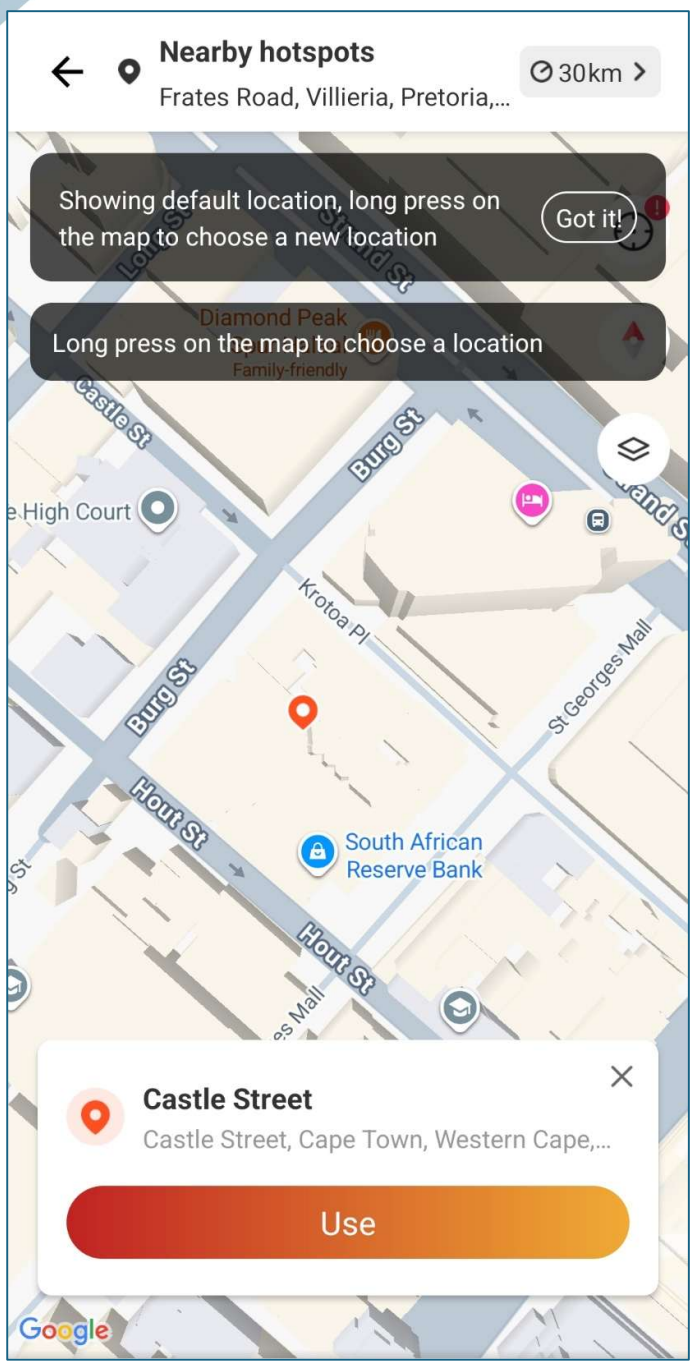


Figure 31 Picture Bird Location setting interface

To chose a location manually, you move the map to where you are (lets use the new campus as the location) and long press (tap and hold) on that spot. Then it will ask you to confirm that the location is correct before using it.

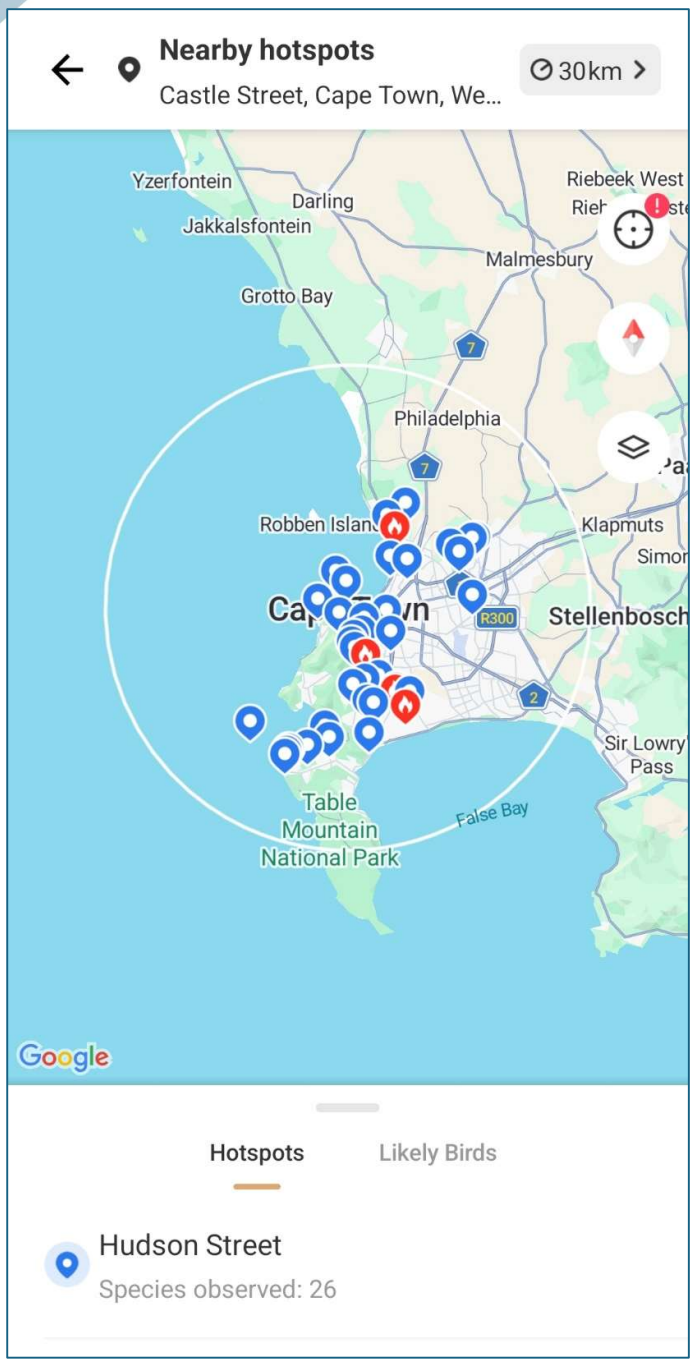


Figure 32 Picture Bird Nearby hotspots

When confirmed, the map will zoom out and show you (within an adjustable radius) all the locations that have been getting a lot of sightings of different birds. The blue is for a good amount of sightings and the red is for "hotspots" where there's loads of people snapping pics and IDing birds from that area. On that note, you'd think that table mountain would have more hotspots.

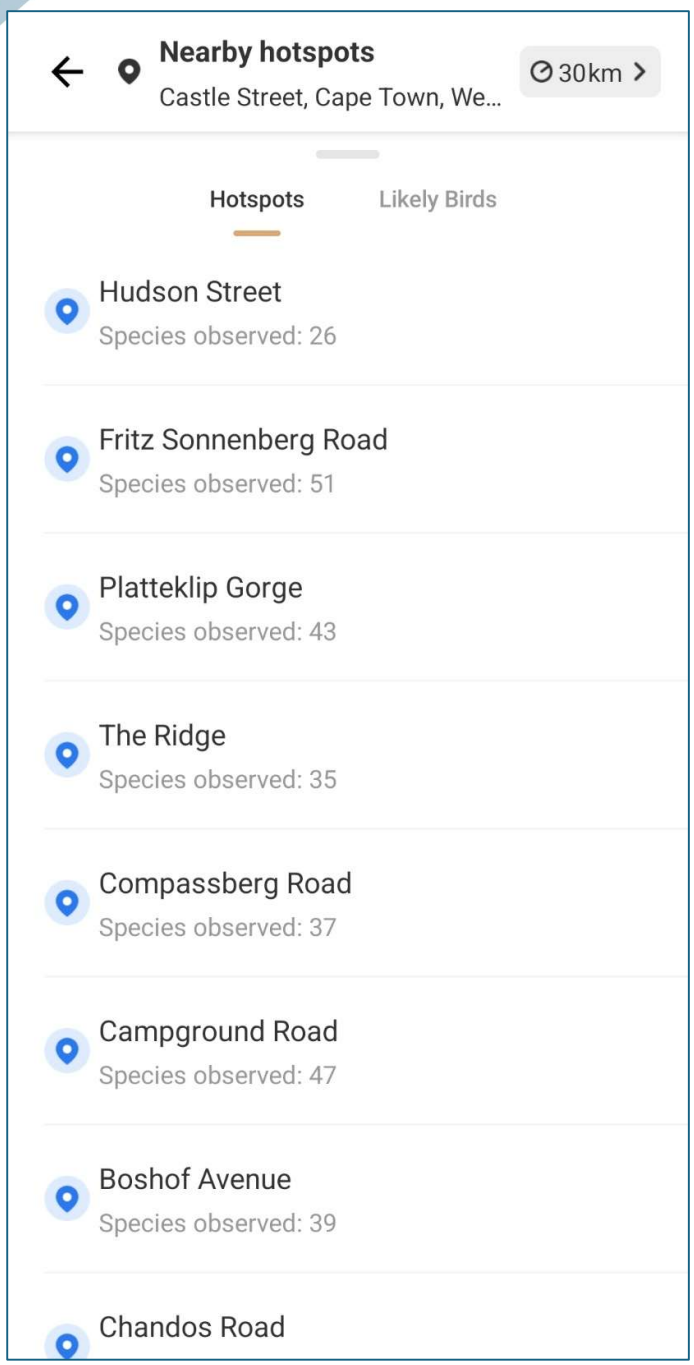


Figure 33 Picture Bird Hotspot listing

You can swipe up to see a list of hotspots instead of using the map. You can chose any spot and it will zoom the map into that spot for you. It also says on the list the number of species (not individual birds, the number of *species*) reported in that location.

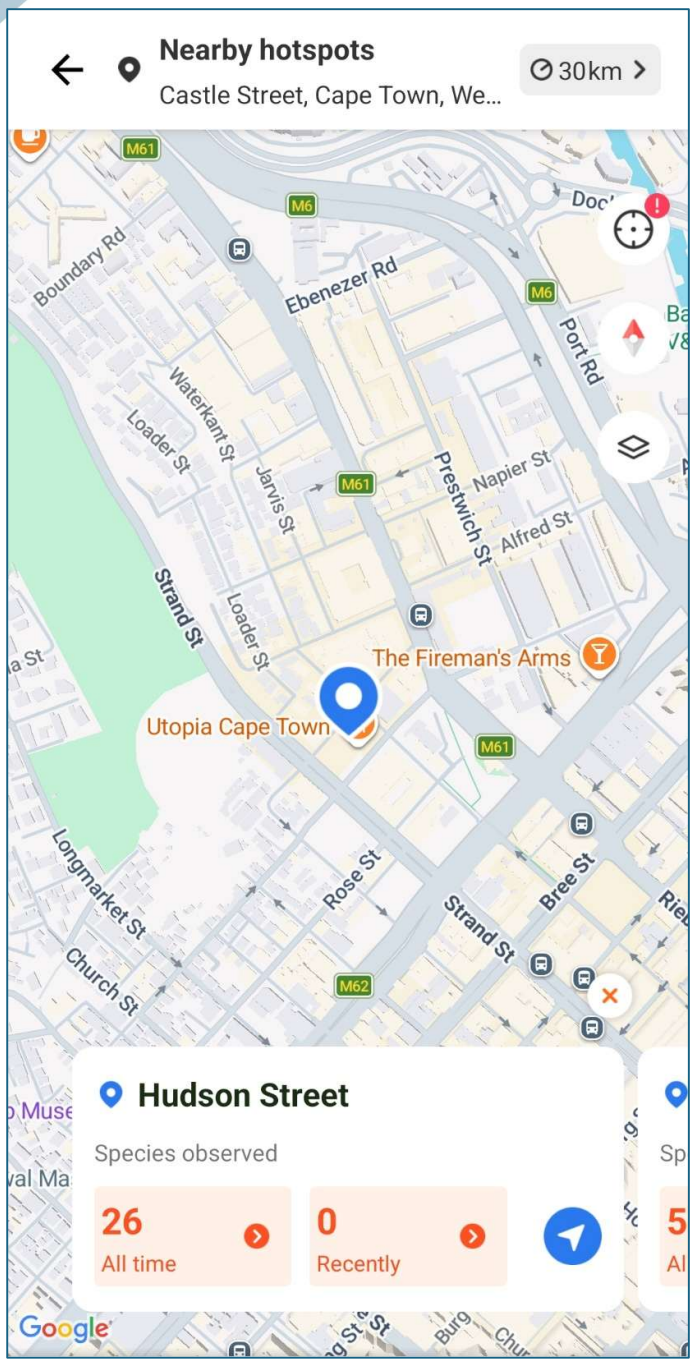


Figure 34 Picture Bird Single hotspot focus

When you tap a listing, here is the zoom in that happens. You can see at the bottom of the screen the name of the location and the count of the number of species seen there overall as well as the recent sightings. You can also get directions to that location with the blue button. As you swipe the bottom ribbon, it will teleport to that spot on the map.

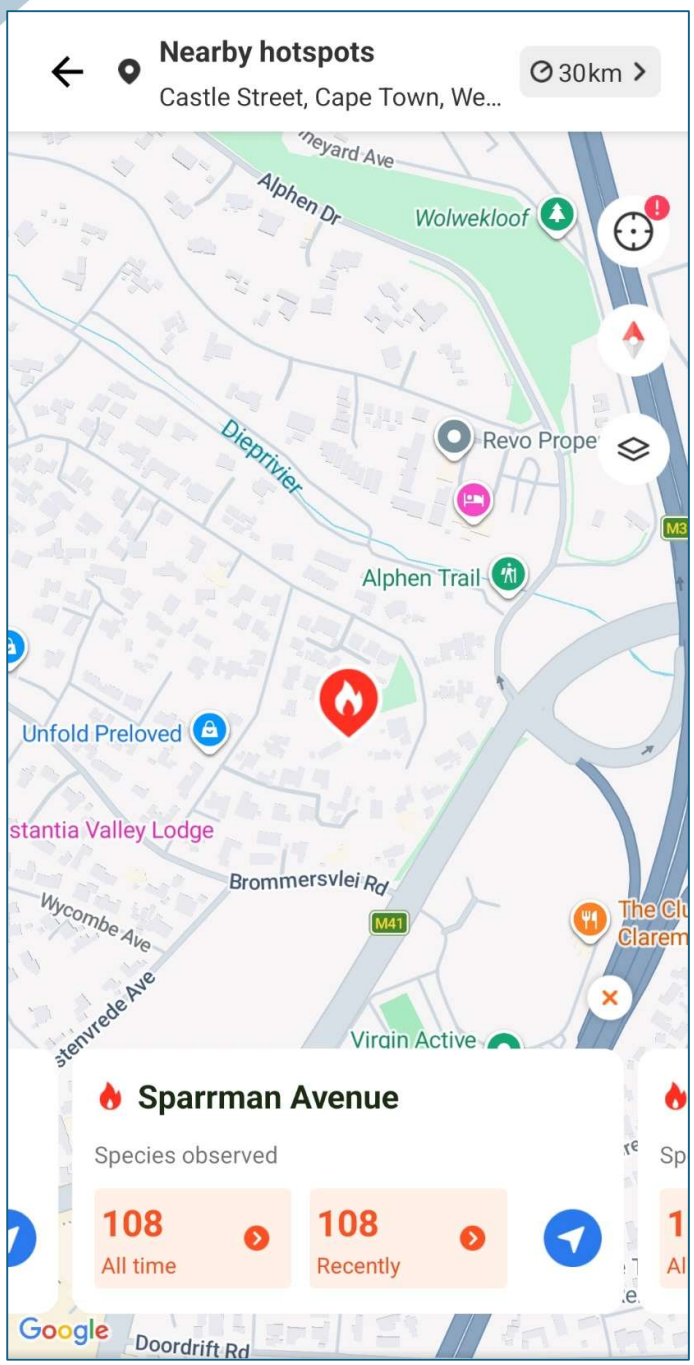


Figure 35 Picture Bird Active hotspot

This is what a real hotspot looks like. This is a spot that gets loads of sightings of many different species very often. Personally, I would like to visit this someday out of curiosity, 108 species is a very sizable amount.

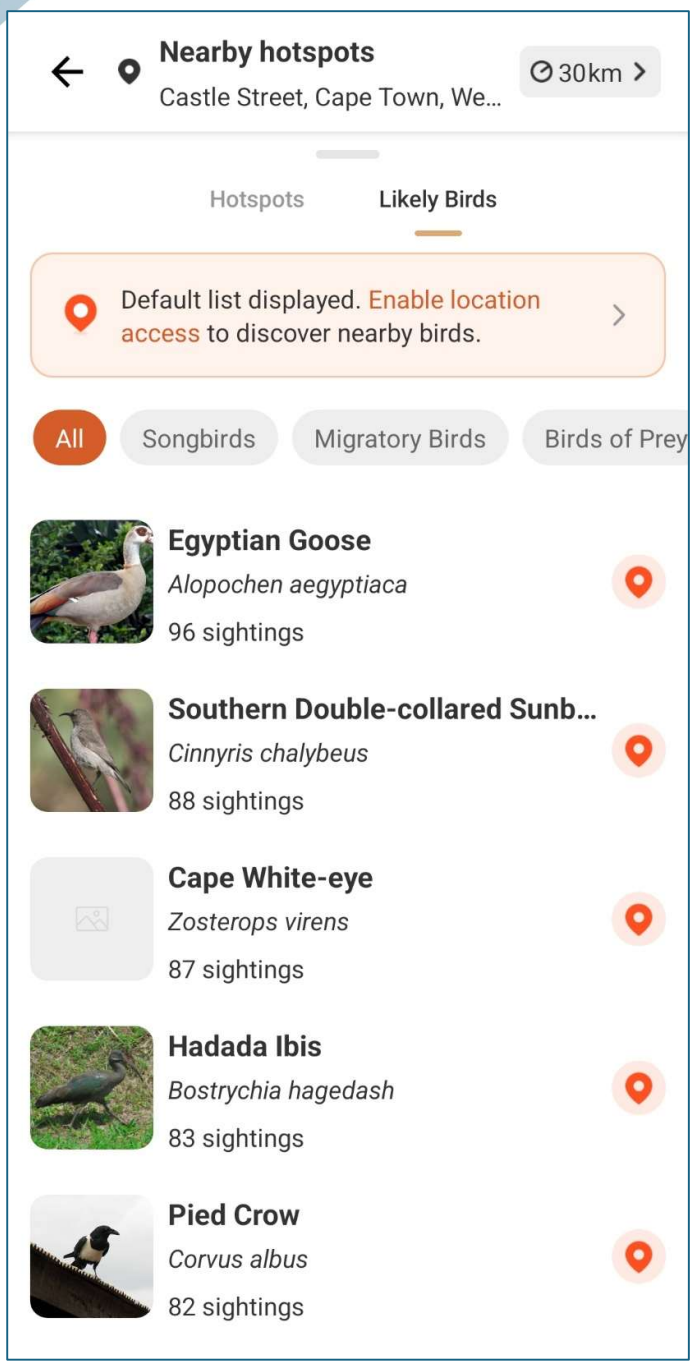


Figure 36 Picture Bird Nearby birds listing

This is the listing of most likely sightings for the area you are in. It is listed in order of reported sightings. I'm not sure why the Cape White Eye doesn't have an image, maybe a small bug in the system.

My application – Pin-a-Bug

Overview

It will essentially be an app that you can open, take a picture of an insect you see nearby, and post the insects picture and its name/species which will then show as a pin on a real time map to let others know of the bugs location. I imagine it as Pokémon Go but for autistic people (as if Pokémon isn't), where you can open the app and see if there are any cool bugs near you to check out. This would be especially cool for rare or migratory insects, as these are often already being documented for research. It would also be useful for monitoring invasive bugs. Mostly it would just be to satisfy the childhood urge many of us still harbour to yell "Yo check out this weird bug!" to all around so you can all gawk at the cool bug together. Other users would be able to interact with the sighting by possibly leaving a reaction of sorts that means "Yup, I saw it too" to make it more interesting and communal. Sightings would probably be taken down after an hour or so as bugs tend to move around, possibly I could include a timed progress bar to let users know how likely it is that the bug is still there that can reset or increase based on the number of "Yup, I saw it" reactions it gets and how recently it got them. If I had the funding I would include an identifying AI or something to identify the bugs as you post, but I do not possess such riches. Perhaps if the Picture Insect developers are willing to hear me out, I could work with them to integrate my little project into their existing apps ecosystem.

Strengths

- The app would be good at getting accurate geolocation data (location pins)
- Users would be anonymous to other users, protecting against people attempting to track users.

Weaknesses

- Not the most secure as I lack funding for real security testing
- May be buggy (heh) due to lack of experience and proper testing teams
- Image filtering (so people don't post nudes or something) will be very primitive, mostly relying on users behaving (will include a report button just in case)

Implementation

I haven't started it yet, but I can assure you it will be implemented in my unique kind of development life cycle. Work will happen when it happens, and commits will vary from one minor bug fix to "I made like 5 new features and forgot to commit them separately oopsies lol", but it will be genuine. There is the possibility for using AI to aid in my coding, but it will be restrained to building only what I instruct it to, I am the creative lead with no exceptions.

Screenshots

View the planning and design document for a look at the concepts

Comparison

Since the apps are all very similar its hard to compare them. The Picture Insect/Bird apps look and mostly function identically as they are made by the same people. My app is smaller scale and there's not much to compare that would be genuinely fair. Theres no real benefit to comparing to features that are impossible to implement at my scale or that aren't relevant to the scope.

	Picture Insect	Picture Bird	My App
Has a map service	No	Yes	Yes
Allows Image Uploading	Yes	Yes	Yes
AI species identification	Yes	Yes	No
Anonymity	Yes	Yes	Yes

Planned Features

While I cannot guarantee that I will be able to implement all that I promise in this Part 1, but I will certainly try.

From Picture Insect, I would take *inspiration* from:

- Elements of their UI
- Their camera functionalities (flash, zoom, framing borders)

From Picture Bird, I would take *inspiration* from:

- Elements of their UI
- Their camera functionalities (flash, zoom, framing borders)
- Their map feature
- The pin system on the map screen
- Filtering map pins based on type of subject (species or general category)

Conclusion

The Picture Insect/Bird apps offer some good pointers on what an app like mine should take into account. While both apps are very similar and have very similar strengths and weaknesses, I am used to using them and such I have a general feel and user perspective that one would not get from writing or crash testing just to do a write up.

My idea is admittedly more grand than I usually keep my projects. I am not the best with Kotlin and Android Studio is not my favourite IDE, but there are guides and people who can help development. While I am usually very aware of the 3 other modules and hence 3 other full on projects I will have to juggle and try to keep things small and simple because of it, since it is a solo project I feel like taking a small risk of making something bigger is calculated. I am, however, notoriously bad at math.

The map UI from Picture Bird gives good insight into what a user can expect from an app like the one I plan to make. The hotspot feature is something that I would like to implement but I am unsure I could pull it off.

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